

Does Easing Migration Barriers Increase Labor Unrest?

Evidence from the Hukou Reform in China*

Nghiêm Q. Huỳnh

Firat Demir

Chenghao Hu

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Abstract

Granting migrants formal legal status is widely thought to reduce crime and social unrest. We find the opposite. China's 2014 Hukou reform eased internal migration barriers most extensively in cities below one million residents. Labor unrest in those cities rose by roughly 30 percent relative to partially reformed cities, concentrated in manufacturing, construction, and transportation, the sectors employing most migrants. Separately collected data on political protests show no comparable increase. Manufacturing employment did not expand, while earnings fell in low-skill services. We argue that formal status expanded migrants' local social and economic entitlements faster than employers and local governments fulfilled them, generating greater contention over jobs and benefits rather than broader political mobilization.

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JEL Codes: J52, J61, J68, O15, R23

*Huỳnh: University of Oklahoma (nqh@ou.edu). Demir: University of Oklahoma (fdemir@ou.edu). Hu: San Francisco State University (chhu@sfsu.edu). Except for the first author, authors are listed alphabetically. We thank Tyler Ransom and participants at the University of Oklahoma 2026 Brown Bag for helpful comments. Firat Demir thanks the Carnegie Corporation of New York for financial support (grant number G-PS-23-60603). The statements made and views expressed are solely the responsibility of the authors.

1 Introduction

Reducing migration barriers is among the most promising policy levers for economic growth,¹ yet the social consequences of such reforms are far less understood. Does easing migration barriers reduce labor unrest? Most evidence on international migrants suggests yes. Granting formal legal status reduces crime and social conflict, with causal evidence from high-income environments where formal employment absorbs new workers at stable wages (Marie and Pinotti, 2024). Whether this pattern extends to internal migration in developing countries, where labor markets face distinct structural constraints, including the persistence of surplus labor, remains an open question. In this paper, we examine this question using China’s 2014 Hukou reform and labor unrest data from 2007–2019 and find evidence that runs counter to the conventional wisdom. Relative to control cities, per capita labor unrest in the cities that received the most extensive barrier reduction rose by 0.106 events per 100,000 residents, a proportional increase of roughly 30 percent in total recorded events.

China’s Hukou system restricts more than 145 million rural workers from full access to public services in their destination cities (Gao et al., 2023). By conditioning access to education, health-care, and social welfare on a worker’s place of birth, the system creates a large population of urban residents without formal residency rights. The 2014 reform relaxed these requirements based on city size, assigning the most extensive barrier reduction to cities below one million residents and only partial reduction to cities between one and five million. We take advantage of this variation in reform intensity for identification in this paper.

The population thresholds in the Hukou reform were set by the central government as part of a national urbanization strategy, rather than in response to local unrest trajectories. Consistent with this institutional setting, pre-reform trends in treatment and control cities track each other closely. Exploiting the Hukou reform variation in a difference-in-differences design, we find that per capita unrest in treatment cities rose sharply after 2014, with the differential widening through 2016 before attenuating. Over the same period, the registered population grew 5–12 percent faster in treatment cities. The sectoral pattern of unrest provides the first piece of evidence on the mechanism. If the effect operates through migrant labor markets, the increase should concentrate in the sectors that employ most migrants. Indeed, unrest rises in manufacturing, construction, and transportation, which together account for more than half of migrant employment nationally, while services show no detectable change.

An alternative interpretation attributes the increase to measurement rather than behavior. The CLB compiles incidents through crowd-sourced electronic collection, and its coverage expanded into smaller cities during the years we study. Recorded unrest could therefore have increased in the treatment cities simply because incidents became easier to detect. We test this possibility using

¹Empirical estimates suggest that eliminating international migration barriers could raise global GDP by 50–150 percent (Clemens, 2011), while easing internal barriers can increase labor productivity by more than 20 percent (Bryan and Morten, 2019). In China specifically, declining internal trade and migration costs account for a substantial share of aggregate labor productivity growth (Tombe and Zhu, 2019).

an independent source. [Qin et al. \(2024\)](#) compile protest events from Radio Free Asia (RFA), drawing on its own correspondents as well as Chinese and international news reports; these events are predominantly political rather than labor-related. The RFA series shows no comparable increase in treatment cities. If improved detection were driving the CLB result, it should plausibly affect both series, yet only labor unrest exhibits the treatment differential. The result also persists when we exclude all transportation incidents, ruling out driver protests associated with the contemporaneous expansion of ride-hailing platforms. Across all five city-size categories, the strike differential follows the reform-intensity gradient in every post-reform year. More generally, changes in incident reporting cannot readily account for increases concentrated in migrant-intensive sectors while leaving incidents in services unchanged.

The key insight is that the reform changed workers' formal status faster than their economic conditions. Formal residency was extended to migrants already working in treatment cities and became easier to obtain for new arrivals. In both cases, it carried greater access to local social insurance and public services and a more permanent stake in the destination city, potentially raising workers' expectations of employers and local governments. The 'registered' population increased by 5–12 percent, while the physical population showed no detectable change ([Lai and Qiu, 2025](#)), indicating that the dominant margin was the formalization of workers already present. However, the material conditions associated with these new rights did not improve immediately. Manufacturing employment showed no differential growth after the reform, while service employment expanded only in a single low-barrier subsector, transportation, where piece-rate earnings declined as more workers entered.² Wage arrears were the dominant recorded grievance after the reform, appearing in 78 percent of events with an identified cause. We therefore examine three reinforcing channels: a gap between newly expanded entitlements and their delivery, congestion in the sectors where migrants work, and a lower cost of collective action under permanent residency. The mechanism section evaluates these channels using evidence on employment, wages, and fiscal capacity.

These findings suggest that the sectoral composition of local labor markets, rather than aggregate economic growth alone, shapes the social consequences of migration reform. When formal status expands faster than migrants' integration into accessible labor markets and labor demand adjusts slowly, migration reform can increase unrest even amid sustained aggregate growth. [An et al. \(2024\)](#) document the migrant wage compression generated by the same reform. We identify a city-level collective consequence of that compression: unrest increased where formalized workers entered sectors that did not expand sufficiently to absorb them. A related concern may arise in other household-registration systems, including Vietnam, where formal registration status similarly determines access to local public services ([Huỳnh, 2025](#)).

²Piece-rate earnings in sectors such as transportation refer to contractual arrangements under which workers (i.e., drivers) are paid per trip or per delivery.

Related Literature. A large body of research has documented significant internal migration costs that limit workers' ability to realize productivity and welfare gains from moving, including institutional barriers that restrict Hukou-registered workers in China (Ngai et al., 2019; Fan, 2019; Hsu and Ma, 2021; Wang et al., 2021; Jin and Zhang, 2023) and spatial frictions that reduce labor mobility more broadly (Imbert and Papp, 2020; Bryan and Morten, 2019). At the firm level, Imbert et al. (2022) show that large Chinese firms absorb internal migrants through expansion rather than wage adjustment, a channel that may not operate in the smaller cities where the 2014 reform had its deepest effect. Whether reducing these frictions affects social stability in destination cities is a different and less studied question. Most existing evidence on migration and conflict comes from contexts where migration is driven by displacement or economic shocks, in which observed social tensions may reflect the forced nature of inflows rather than the effects of policy itself. Even when governments deliberately ease internal migration barriers, reforms either apply uniformly across regions, leaving no variation to identify causal effects, or operate through infrastructure investment that simultaneously changes trade costs and market access, confounding the migration channel.

The 2014 Hukou reform in China provides a distinct setting for studying the consequences of changes in formal status. The reform changed administrative residency requirements without altering physical infrastructure or trade costs, while creating a gradient of reform intensity across cities of different population sizes. Previous research using natural experiments finds that granting formal legal status to international migrants reduces individual crime (Pinotti, 2017; Mastrobuii and Pinotti, 2015; Bell et al., 2013; Marie and Pinotti, 2024). In those settings, legal status primarily operates through employment: it removes restrictions on formal work and raises the returns to legitimate employment. Consistent with this mechanism, removing immigration restrictions in Switzerland increased firm performance and worker productivity as local labor markets absorbed new workers through firm creation and wage adjustment (Beerli et al., 2021). The Hukou system operated differently. Migrants already participated in local labor markets regardless of their Hukou status. Formal residency instead changed what they could claim in the destination city—including access to public services, social insurance, and a more permanent right to stay. Moreover, the outcome here is collective labor action rather than individual crime.

These differences suggest why formalization need not have the same consequences across different settings. When labor demand is sufficiently strong, employers may have incentives to accommodate the conditions associated with formal status, allowing workers to gain both status and improved economic conditions. This is the case in Pinotti (2017) and Marie and Pinotti (2024). When labor supply exceeds the number of jobs accessible to migrants, employers face less pressure to improve working conditions. Wages can fall in flexible-pay sectors, while workers face queues or limited access in sectors where employment is constrained, widening the gap between formal entitlements and actual economic conditions. The 2014 Hukou reform occurred in cities where these labor-market constraints were particularly strong. An et al. (2024) show that migrant wages fell by 2.6–7.9 percent on average in non-megacities after the reform, with declines increas-

ing monotonically as city size fell. We document the corresponding city-level increase in labor unrest that followed the reform.

Although granting legal status can reduce individual crime, a separate literature finds that migration inflows can increase collective conflict when receiving areas have limited capacity to absorb them. [Helms \(2024\)](#) finds that a 10 percentage point increase in migrant inflows, induced by India’s trade exposure, increased the incidence of anti-migrant riots by 6.5 percentage points. [Knight and Tribin \(2023\)](#) show that Venezuelan displacement to Colombian border municipalities increased homicide rates, driven by the victimization of migrants rather than crimes they committed. [McGuirk and Nunn \(2024\)](#) provide the closest parallel to the mismatch mechanism we study: agricultural development projects in traditionally pastoral regions of Africa nearly doubled the risk of conflict when local economies could not absorb the resulting disruption. These studies and ours point to a common possibility: conflict can emerge when migration inflows or policy changes—and the pace at which they occur—outstrip the capacity of receiving environments to absorb them.

Our paper differs from these studies in both the type of migration and the type of conflict examined. The conflicts they document arise from anti-migrant backlash, migrant vulnerability, or competition over resources following migration driven by forces outside the receiving government’s control. We instead study collective labor conflict generated by a targeted government policy that expanded migrants’ access to formal residency, with unrest concentrated in manufacturing, construction, and transportation. Within China, previous studies examined the structural conditions and state responses that shape collective unrest ([Elfstrom and Kuruvilla, 2014](#); [Cai and Chen, 2022](#); [Lorentzen et al., 2013](#); [Qin et al., 2024](#); [Wen, 2025](#)). Closest to our labor-unrest outcome, [Campante et al. \(2023\)](#) show that the mid-2010s export slowdown increased CLB-recorded strikes in trade-exposed prefectures and that the central government replaced local leaders whose strike management lagged behind peers facing comparable shocks. Their mechanism begins with a temporary market contraction operating through layoffs and wage arrears; ours begins with a permanent administrative reform that changed workers’ formal status. Controlling directly for trade exposure leaves our estimates unchanged.

We are not the first to study the relationship between the Hukou reform and unrest. Closest to our study, [Lai and Qiu \(2025\)](#) examine the 2014 reform and report lower unrest using a regression discontinuity design at the 3 million population threshold. Our approach differs in three ways that strengthen the identification and reveal dimensions of the reform that their design does not capture. First, we estimate the average treatment effect across cities where the reform eliminated or substantially reduced migration requirements, rather than a local effect at the 3 million threshold comparing partial reform with no reform. Second, we exploit a five-group reform-intensity gradient that allows treatment effects to vary with the extent of reform, including a strike differential that follows the reform-intensity ordering in every post-reform year. Third, our panel contains seven years of pre-reform data, allowing a more extensive assessment of pre-treatment trends. Section 3.2 examines these differences in detail. On the reform’s other margin, [Guo et al.](#)

(2025) use linked employer-employee data to show that tighter employer-sponsored Hukou quotas increased migrants' relative wages, further indicating that Hukou status has economic value within the employment relationship.

The rest of the paper is organized as follows. Section 2 presents the research design and data. Section 3 reports the main results. Section 4 investigates mechanisms. Section 5 concludes.

2 Research Design

The central challenge in identifying how easing migration barriers affects labor unrest is separating the reform's direct impact from other forces that shape city-level labor conditions over the same period. The 2014 reform's city-size classification provides a credible source of identification because the population thresholds determining reform intensity were set centrally as part of a national urbanization strategy, announced only in July 2014 ([State Council of the People's Republic of China, 2014b](#)), and bear no direct connection to pre-existing unrest patterns at the city level.

2.1 The 2014 Hukou Reform

In July 2014, the Chinese government announced a landmark reform of the household registration system (Hukou) that had shaped China's social and economic structure for over half a century. Originally introduced in the 1950s, the Hukou system tied individuals to their place of birth, classifying them as rural or urban residents and determining their access to education, healthcare, housing, and social welfare. The system created stark divisions between rural and urban populations while limiting the mobility of hundreds of millions of rural migrants seeking better opportunities in cities.

The 2014 reform, implemented through the "National New-Type Urbanization Plan (2014–2020)" and related State Council directives, aimed to encourage rural migrants to gain formal residency status in urban areas as part of China's broader urbanization strategy. The reform expanded the eligibility of migrants for local public services, including education, social security, healthcare, and public housing ([Zhang and Treiman, 2013](#); [Wu and Treiman, 2004](#)), narrowing the welfare gap between migrants and registered urban residents ([Xie and Zhou, 2014](#)).

The 2014 reform classified Chinese cities into five categories based on the size of their urban population, each implementing differentiated approaches to reducing Hukou requirements ([State Council of the People's Republic of China, 2014b](#); [An et al., 2024](#)). Cities with populations less than 0.5 million removed their administrative restrictions, leaving legal stable residence, rental included, sufficient to register. Medium-sized cities (0.5–1 million) experienced substantial reduction in requirements, with migrants needing one to three years of social security contributions along with stable employment. Larger cities (1–3 million residents) maintained controlled issuance, requiring three to five years of contributions, while cities with 3–5 million residents imposed even stricter criteria. Megacities such as Beijing and Shanghai, with populations exceeding 5 million, maintained strict controls with no significant reduction in requirements. In our em-

pirical analysis, we map these five categories to a binary assignment. Cities in the two smallest categories (populations below 1 million, sizes 4 and 5) form the treatment group, which received the most extensive reduction in Hukou requirements. Cities in the two intermediate categories (populations between 1 and 5 million, sizes 2 and 3) form the control group, which experienced only partial barrier reduction. We use “treatment” and “control” throughout.

The Hukou registration confers permanent formal residency in the destination city, entitling holders to local public services, education, and social insurance. The reform therefore targeted a barrier different from the temporary residence permits that govern short-term labor mobility. The population response we document reflects permanent or long-term in-migration rather than circular flows.

2.2 Measurement of Labor Unrest

We obtained the data on labor unrest, including labor strikes and protests, from the China Labor Bulletin (CLB; [China Labor Bulletin, 2024](#)), a Hong Kong-based non-governmental organization dedicated to supporting the development of trade unions within China. This dataset is widely used in research on labor unrest, labor rights, and state-society relations ([Qin et al., 2024](#)) and is considered one of the most comprehensive sources of information on strike and protest events in China. Since 2007, the CLB has systematically collected labor strike and social protest statistics, with electronic coverage available from 2011 to date.³ The CLB data measure reported incidents of labor unrest, including labor strikes, protests, demonstrations, and other forms of collective action. Each record identifies the timing, prefecture-level city location, employer name and description, industry classification, number of participants, worker actions, and government responses. The CLB data compilation draws from multiple sources, including overseas Chinese media, labor activists within China, and internet searches that cover social media platforms.

Our analysis examines labor unrest events from 2007 to 2019, a period longer than comparable studies in this literature. The sample begins in 2007, when the CLB commenced systematic data collection, and ends in 2019 to avoid confounding from COVID-19 lockdowns and concurrent changes in internet censorship enforcement that disrupted both labor market conditions and the CLB’s collection methods after 2019.⁴ The CLB recorded roughly 12,700 unrest events during this time span. We split events by the recorded worker action. Actions whose description contains “strike” are coded as strikes, and the remaining collective actions, including demonstrations, sit-ins, road blockages, petitions, demands for government intervention, and threats to jump, are coded as protests. For 2011–2019, strike takes priority when an event records both strike and protest actions, as about 6 percent do. In the 2007–2010 data, the strike and protest flags can

³For strike data prior to 2011, similar information was extracted from annual CLB reports and supplemented with additional sources such as Boxun, a U.S.-based political news website targeting Chinese audiences. Coverage of the smallest cities was particularly sparse before 2011, with between zero and five total events per year across all 51 size-5 cities in 2007–2010.

⁴CLB coverage in 2009 sits well below the adjacent years, at 35 recorded incidents against 69 to 90 in 2007, 2008 and 2010 (Appendix Table A1). The treatment-control differential in 2009 is the smallest of any pre-reform year, but excluding it only moves the pooled estimate from 0.106 to 0.107.

overlap, so the two category counts do not form a mutually exclusive partition. Under these rules, the data contain about 9,400 protest events and 3,200 strikes. 78 percent of incidents involve fewer than 100 participants. The CLB began recording worker grievances in 2011, and wage arrears are the leading cause, named in 74 percent of events with a recorded grievance over 2011 to 2019.

In addition to the CLB, we rely on city-level data from the China Statistical Yearbook ([National Bureau of Statistics of China, 2019](#)). We use the 2013 registered population of the municipal districts to classify cities into size categories, and annual registered population for per capita rates, employment, and fiscal variables for the mechanism analysis.⁵ Our final dataset includes 285 prefecture-level cities, providing coverage comparable to most existing studies examining the 2014 Hukou reform. Throughout the paper, “city” refers to a prefecture-level city, China’s principal subnational administrative unit below the province. All fixed effects, clustering, and treatment assignments operate at this level. The sample contains 285 cities distributed across the five size categories, with 12 megacities (Category 1), 11 large cities (Category 2), 108 medium cities (Category 3), 103 small cities (Category 4), and 51 size-5 prefecture-level cities (Category 5).

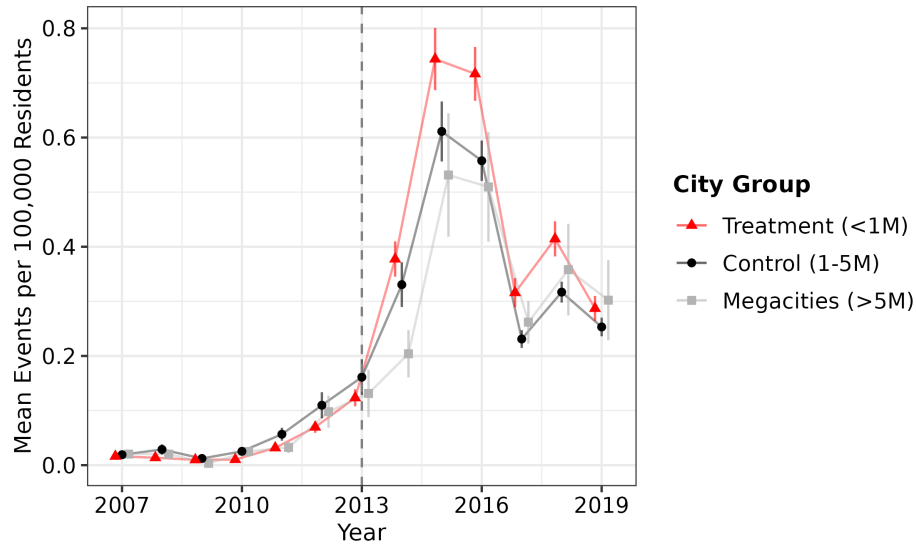
We construct three outcome variables from these events. *Labor Unrest* is the total count of incidents in a city in a given year t , while *Strike* and *Protest* count the two categories defined above. This disaggregation helps us distinguish work-stoppage actions from other forms of collective labor grievance, which may respond differently to the Hukou reform.

For each of these three outcomes, we examine two complementary measures, the raw event count and the per capita rate (events per 100,000 registered population). The appropriate measure of labor unrest exposure is incidents per resident, not total incident volume. A city of five million with ten strikes and a city of 500,000 with ten strikes present identical events to a count-based measure, yet residents of the smaller city experience ten times the disruption per capita. Per capita rates are particularly relevant here because the reform itself induced differential population growth across city groups. When differential population growth alters the denominator, raw counts that appear comparable across groups can mask divergent per capita rates. Per capita unrest rose because incidents grew faster than the expanding registered population, a divergence our raw counts understate. Therefore, we treat per capita rates as the primary outcome variable. However, the per capita unrest rates are right-skewed, with many city-year observations equal to zero. The OLS estimation imposes a linear conditional mean on a non-negative outcome with this shape. Therefore, we also repeat our benchmark estimations using PPML, which requires no linearity assumption for a non-negative outcome.

CLB coverage expands over the sample period. Appendix Table A1 documents total incident

⁵Registered population is the yearbook’s year-end Hukou-registered population of the municipal districts (shixi-aqu), averaging 1.4 million in 2013. The districts are the urban core, so the geography matches the area the reform’s thresholds refer to. The thresholds themselves are stated in resident population, which the classification standard defines to include migrants without local Hukou ([State Council of the People’s Republic of China, 2014a](#)), so a city near a threshold with many migrants may fall into a lower size category under our measure than under the reform’s. The yearbooks also report registered population for the whole prefecture, averaging 4.4 million, which we do not use because the reform’s thresholds and the labor markets we study operate at the city level. Cities with incomplete population data are excluded and account for a minor share of national economic activity.

Figure 1: Mean Per Capita Labor Unrest by City Group (2007–2019)



Notes: The figure plots mean per capita labor unrest (strikes and protests combined, events per 100,000 registered population) by year for treatment cities (population below 1 million, sizes 4–5), control cities (population 1–5 million, sizes 2–3), and megacities (population above 5 million, size 1). The dashed vertical line marks 2013, the last pre-reform year.

counts by year and city size. The CLB recorded fewer than 100 events per year before 2011, rising to over 600 in 2013 and peaking at roughly 2,600 in 2015–2016 before settling near 1,300 by 2019. Part of that growth is a real increase in conflict and part is a change in how the CLB documents it. The organization moved to electronic crowd-sourced collection after 2011, social media reporting expanded across China over the same years, and the CLB has acknowledged that its sampling rate varied over time, with earlier coverage concentrated in Guangdong and major cities.⁶

Despite the growth in coverage, external validation indicates that the series tracks real conflict. [Campante et al. \(2023\)](#), who use the same records, cross-check them against the official count of labor-dispute arbitration cases from the China Labour Statistical Yearbooks. The two series move together in levels and in changes, and the ratio of CLB to official cases is uncorrelated with local economic shocks, so what the CLB captures is relative unrest across time and place, not the intensity of its own collection effort.

The CLB data are therefore informative on where and when unrest rose. Figure 1 displays the mean per capita labor unrest from 2007 to 2019 for the three city groups, treatment cities below one million residents, where the reform cut the deepest, control cities of one to five million, which saw only partial relaxation, and megacities above five million, which saw none. We observe two patterns. Every group rises sharply after 2013, and treatment cities rise the most. The mean unrest per 100,000 goes from 0.039 to 0.476 in treatment cities, from 0.059 to 0.383 in control cities, and from 0.047 to 0.361 in megacities. Treatment cities begin lowest and end highest, whether we read

⁶See “China’s Labour Movement in Transition,” *Made in China Journal*, July 2018, <https://madeinchinajournal.com/2018/07/07/chinas-labour-movement-in-transition/>.

the change as a difference, 0.436 against 0.324 and 0.314, or as a multiple, 12.1 against 6.5 and 7.6. Through 2013 the three series rise together as CLB coverage expands, with treatment cities lowest throughout, and from 2014 the ordering reverses.

The timing lines up with the reform, and the ordering across groups matches its intensity. The pattern is not, however, evidence that the Hukou reform caused the increase, because cities differ in much besides their exposure to the reform, and those differences moved over the same years. This is what we examine next.

2.3 Identification Strategy

We adopted a difference-in-differences event study design to estimate how variation in the intensity of Hukou barrier reduction across city types affected labor unrest. The design compares cities that received the most extensive reduction in migration barriers (the treatment group) to cities where restrictions were only modestly eased (the control group). Treatment cities are those with 2013 registered populations below 1 million (sizes 4 and 5 in the reform’s classification scheme), where the 2014 policy mandated complete or near-complete elimination of Hukou requirements. Control cities have populations between 1 and 5 million (sizes 2 and 3), which experienced only partial barrier reduction through requirements such as multi-year social security contributions or stable employment thresholds.

We used the 2013 registered population for classification to avoid potential endogeneity from cities adjusting reported population in response to the reform. Because the specific population thresholds were announced only in July 2014, the pre-announcement population reflects predetermined city characteristics rather than strategic responses to the policy change.⁷ The treatment group includes 154 cities and the control group includes 119 cities, yielding 3,510 city-year observations over the 2007–2019 panel. Appendix Table A2 reports pre-reform sample characteristics. Treatment cities are smaller by construction, since treatment is assigned by city size, and have correspondingly lower population, employment, and fiscal levels. Pre-reform per capita unrest rates are modestly lower in treatment cities (0.04 versus 0.06 per 100,000). These level differences are absorbed by city fixed effects and do not threaten identification. The key diagnostic is the parallel pre-reform trends assumption, which we examine in Section 3.

We conduct the baseline comparison between cities with full and partial reductions in migration barriers, setting aside megacities (populations above 5 million, size 1) for now. The rationale is comparability. Comparing the two reform groups keeps the broader policy environment more similar and isolates variation in reform intensity. Megacities did not receive any reduction in migration barriers, differ from smaller cities in political status and economic structure, and number only 12 in our sample. Their financial-center intensive economies were also exposed to the mid-2015 stock-market crash and the anti-corruption campaign in 2015 in ways that smaller cities were

⁷As a robustness check, we verify that results are quantitatively similar when using 2014 population for classification (Appendix Table A3). Only 17 of 285 cities (6 percent) change size category between 2013 and 2014, confirming that the classification is stable.

not. The raw means in Figure 1 peak in 2015 at 0.74 events per 100,000 in treatment cities, 0.61 in control cities, and 0.53 in megacities. Thus, identification does not rely on megacities being uniquely exposed to these shocks; the difference-in-differences design nets out the common rise in unrest. Section 3.4 shows that the strike results are robust to using megacities as the comparison group.

The binary contrast is also a deliberate estimation choice. The reform assigns five intensity levels, and one could instead model unrest as a function of a multi-valued or continuous treatment. However, when treatment effects vary between intensity levels, the coefficient on such a treatment does not have a clear interpretation, because it aggregates the level-specific effects with weights that can be negative (Callaway et al., 2024). The binary comparison of the most- and least-treated cities identifies a transparent average effect under the standard parallel-trends assumption. The cost is that it pools across intermediate intensity levels, which we address by reporting the full five-group responses in Section 3.

We estimate the following event-study specification in Eq. (1):

$$y_{it} = \sum_{k=-6, k \neq 0}^6 \beta_k \times \text{Treat}_i \times \mathbf{1}(t = 2013 + k) + \alpha_i + \gamma_{p(i)t} + \varepsilon_{it} \quad (1)$$

where y_{it} represents the outcome of interest in city i and year t (2007–2019), therefore each observation is at the city-year level. The treatment indicator, Treat_i , equals one for treatment cities (2013 population below 1 million) and zero for control cities (population between 1 and 5 million). The event-time coefficients β_k capture the differential change in outcomes for treatment cities relative to control cities in year $2013 + k$ compared to the omitted reference year 2013, with k ranging from -6 (year 2007) to 6 (year 2019). The pre-reform years are 2007 through 2012. The city fixed effects, α_i , absorb time-invariant city characteristics, while $\gamma_{p(i)t}$ denotes province-by-year fixed effects for province $p(i)$ of city i , controlling for all time-varying shocks at the province level, including common trends, provincial economic conditions, and concurrent reforms. Standard errors are clustered at the city level to account for arbitrary serial correlation within cities.

Our estimation design identifies the average treatment effect of reform intensity on city-level per capita unrest, treating the one million population threshold as the assignment rule to the treatment group. The dependent variable equals labor unrest events per 100,000 registered (Hukou) population. Because the registered population itself is an administrative outcome of the reform, the rate measures unrest per ‘legal’ resident rather than per ‘physical’ resident. This measure is the policy-relevant quantity, as local governments allocate services and enforce labor standards on the basis of registered population.

The design relies on three assumptions. The first is parallel trends, which requires that treatment and control cities would have followed parallel outcome paths in the absence of the differential Hukou reform. The pre-reform event study coefficients provide a direct test of this assumption. For the primary per-capita OLS outcomes, joint tests do not reject zero pre-reform coefficients for overall unrest ($p = 0.619$), strikes ($p = 0.665$), or protests ($p = 0.744$). The count PPML ro-

bustness tests likewise do not reject ($p = 0.404, 0.403, \text{ and } 0.866$). Appendix Table A4 reports both sets of tests, and Section 3 examines the outcome paths directly. Flat pre-trends are necessary but not sufficient for the assumption to hold after the reform. Province-by-year fixed effects absorb all time-varying shocks common to cities within a province. However, they cannot absorb within-province shocks that load differentially on city size, as city size defines treatment assignment. In Section 3.4 we verify robustness to city-specific linear time trends, which additionally control for differential city-level trajectories.

The second assumption is no anticipation, which requires that cities did not differentially adjust their behavior before the reform took effect. The specific population thresholds defining treatment intensity were announced only in the State Council’s July 2014 directive ([State Council of the People’s Republic of China, 2014b](#)), so cities had no basis to anticipate their group assignment in advance. Even if the general direction of Hukou reform was signaled at the 18th Party Congress in November 2012, responding to yet-to-be-announced numerical thresholds would have required cities to anticipate specific cutoffs that had not yet been formalized. We examine whether population trajectories in treatment and control cities diverged before 2014 in Figure 2, and also confirm the robustness of our findings to alternative city-size classification windows in Section 3.4.

A related condition requires that the population thresholds be uncorrelated with pre-existing trends in labor unrest. The thresholds correspond to China’s longstanding national city-size classification system and were determined centrally as a uniform policy, not in response to local unrest trajectories. Any pre-existing level differences between city groups are absorbed by the city fixed effects, and the pre-reform patterns documented in Section 3 show no evidence of differential trends before 2014.

A final concern involves spillovers across cities (Stable Unit Treatment Value Assumption). In fact, [Qin et al. \(2024\)](#) document that protest in one Chinese city increases protest probability in connected cities by 17 percent within two days through social media. Because spillover protests materialize within the same calendar year as the instigating event, annual aggregation fully captures any contamination of control-city counts rather than washing it out. If treatment-city unrest spills into control cities, our DiD estimate is biased toward zero, making the positive estimates a conservative ‘lower bound’ on the true differential effect.

Treatment is collinear with city size, so identification fails for any unobserved post-2014 shock whose incidence varies with size and is not absorbed by the fixed effects. The province-by-year fixed effects absorb shocks common to cities within a province, including provincial policy changes, economic cycles, and enforcement campaigns. They cannot absorb a shock that varies across cities within a province and loads on size. Here, two kinds of such shock matter. The first is economic. The post-2014 export slowdown fell hardest on manufacturing-heavy smaller cities, while the supply-side capacity cuts from 2015 closed coal, steel and cement plants concentrated in industrial cities. The environmental campaign shut factories across northern China, and the anti-corruption campaign disrupted patronage networks where state-owned employment was largest. Although each of these could generate unrest on its own, Section 3.4 controls for them directly,

and the estimates hold. The second is measurement. The CLB finds incidents through crowd-sourced electronic collection, and that collection reached smaller cities over the same years when the reform took effect, so improved detection would mimic the effect we estimate. If detection rather than unrest drove the increase, then a separately collected record of events would show the same rise, and Section 3.4 tests that prediction against data from outside the CLB.

The dose-response pattern documented in Section 3 strengthens the case that the Hukou reform drives our result. The smallest cities show the largest increase in per capita unrest, and for strikes, the four size groups order exactly as reform intensity does in every post-reform year. A confounder of either kind would have to produce not only a post-2014 break at the one million threshold, but that same ordering across the full distribution. We know of no other national program that changes discontinuously at that threshold. Appendix Figure A1 quantifies how large a violation of parallel trends would have to be to overturn the result.

3 Results

We report the results in three parts, followed by a robustness section. Section 3.1 examines registered population, the outcome the reform bears on most directly. Section 3.2 estimates the effect on per capita unrest, and Section 3.3 decomposes it by sector and incident scale. Section 3.4 tests the alternatives.

3.1 Population Response to the Hukou Reform

If the 2014 reform differentially eased migration barriers in treatment cities, its most direct observable consequence should be a differential increase in the registered resident population. We therefore first examine whether the reform produced a corresponding change in the registered population. We estimate Eq. (1) using the log of registered population as the outcome variable, retaining the same city and province-by-year fixed effects, the 2013 reference year, and city-clustered standard errors.

Figure 2 presents event study estimates for (log) registered population, comparing the treatment and control cities. An increase in registered population suggests that more migrants gained formal residency in these cities relative to cities that retained higher Hukou requirements. The pre-reform coefficients are statistically indistinguishable from zero across all years, and a joint F -test that the estimable pre-reform coefficients are zero fails to reject ($F = 1.53$, $p = 0.170$, Appendix Table A4), supporting the parallel trends assumption for population dynamics. After 2014, treatment and control cities diverge. By 2014–2019, treatment cities had approximately 5–12 percent higher registered population than control cities, peaking in 2017 before easing slightly. Each individual post-reform year coefficient is statistically significant at the 5 percent level, though the 2014 estimate ($p = 0.04$) only narrowly clears the threshold. The pooled DiD estimate of 0.057 ($p < 0.10$, Appendix Table A7) averages across the buildup period and is therefore smaller than the peak-year coefficients.

Treatment cities averaged approximately 603,000 registered residents before the reform. Registered population also increased in the control cities, by approximately 19 percent over the post-reform period, implying a counterfactual registered population of roughly 719,000 in treatment cities absent a differential treatment effect. Relative to this counterfactual, the estimated event-study effects imply approximately 36,000–80,000 additional registered residents per treatment city.⁸ The timing and magnitude of this response are consistent with the reform’s greater reduction in migration barriers in treatment cities increasing access to formal residency status, while cities subject to less extensive reform continued to impose greater restrictions on Hukou transfers.

Registered population can increase through two distinct channels: the formalization of migrants already residing in a city and new migration induced by the reform. Only the latter increases the physical population stock of the city. [An et al. \(2024\)](#), using restricted-use 2015 minicensus records, find that after 2014 migration flows shifted toward smaller administrative cities and away from larger ones, while the share of migrants remaining within their original city-size category changed little. This finding documents a change in the direction of relative migration flows, rather than an increase in the total population of treatment cities. Moreover, because the data do not distinguish urban from rural destinations within an administrative city, some of the observed movement may reflect migration back to rural hometowns rather than new entry into urban labor markets. Consistent with this interpretation, [Lai and Qiu \(2025\)](#), using physical urban population stocks from the Urban Construction Statistical Yearbook, find no discernible effect of the reform on urban population. Taken together, the evidence indicates that the reform substantially increased registered population without a corresponding increase in the physical population stock, pointing to formalization as the primary margin of adjustment and leaving, at most, a modest contribution from new arrivals. The next question is whether this expansion of registered population affected labor unrest and, if so, through which channels.

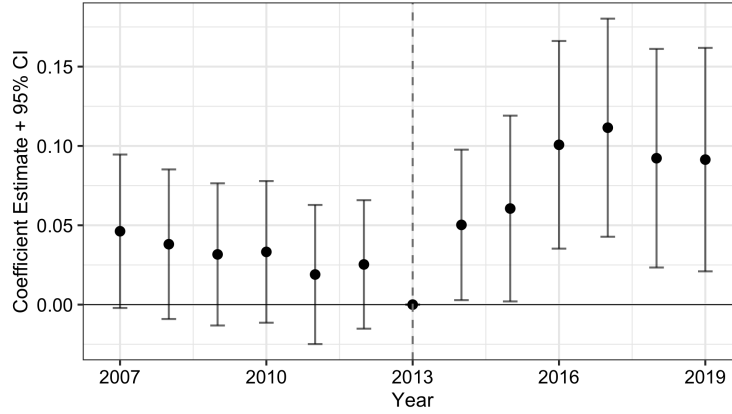
3.2 Per Capita Unrest Rates

In Figure 3 we present the event study estimates of Eq. (1) for per capita unrest rates. The pre-reform coefficients range between -0.004 and $+0.011$ events per 100,000, with p -values from 0.68 to 0.94, and the same pattern holds for strikes and protests, supporting the parallel trends assumption.⁹ After 2014, per capita unrest increased in treatment cities relative to control cities. The overall effect builds from 0.089 events per 100,000 in 2014 to a peak of 0.184 in 2016 ($p < 0.01$), and the post-reform coefficients are jointly significant ($p = 0.021$). Effects remain elevated through 2018 before declining toward zero in 2019, and both strikes and protests are elevated throughout 2014–2018 in panels (b) and (c). The attenuation after 2016 is consistent with labor market adjust-

⁸The pre-reform mean of registered population in treatment cities is 60.30, measured in units of 10,000. The event-study coefficients range from 0.050 in 2014 to 0.111 in 2017, corresponding approximately to 5–12 percent differences in registered population relative to the 2013 baseline. Appendix Table A7 reports the mean of (log) registered population (4.472); the corresponding level mean cannot be recovered directly from this log mean because of Jensen’s inequality.

⁹The parallel trends pattern is robust to adding city-specific linear time trends, with a joint pre-reform test of $p = 0.671$ (Appendix Table A4).

Figure 2: Population Response to the Hukou Reform (2007–2019)



Notes: The figure plots event study coefficient estimates and 95% confidence intervals for log registered population, comparing treatment cities (sizes 4–5, population below 1 million) to control cities (sizes 2–3, population 1–5 million). The pooled DiD estimate is 0.057 (s.e. 0.033, $p < 0.10$, Appendix Table A7). The omitted reference year is 2013. All regressions include city and province-by-year fixed effects with standard errors clustered at the city level. Megacities (size 1, population above 5 million) are excluded.

ment as migrants are sorted into accessible employment over time, a channel we explore further in Section 4.

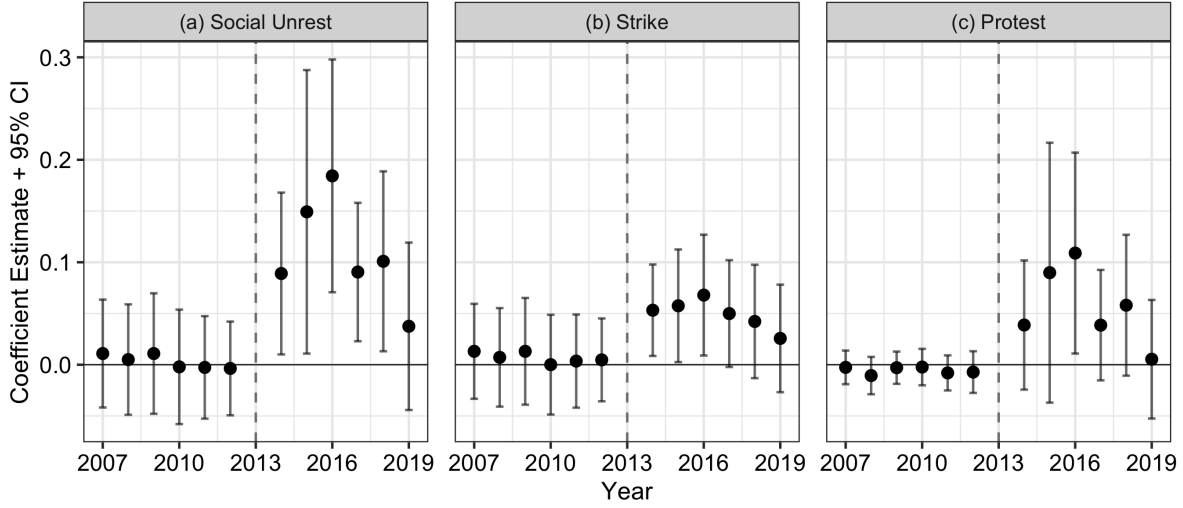
Table 1 collects the pooled difference-in-differences estimates under three measurements of the same data. Columns (1) to (3) report per capita rates, events per 100,000 registered residents, estimated by OLS. The time-varying registered population is the primary and policy-relevant denominator, given that the reform raised registered populations in treatment cities (Section 3.1). Columns (4) to (6) report PPML estimates on raw event counts, which measure proportional changes without a denominator. Columns (7) to (9) repeat the rate specification of columns (1)–(3) with 2010 registered population as a fixed denominator. Column (1) reports a sustained differential of 0.106 events per 100,000 ($p < 0.01$), with 0.043 for strikes ($p < 0.01$) and 0.061 for protests ($p < 0.05$). Set against the control-group post-reform mean of 0.383, that differential is sizable, and the implied proportional effect of roughly 28 percent lines up with the 29.3 percent from the count data in column (4).¹⁰ Total recorded events rose by 29.3 percent ($p < 0.05$, column 4) and strikes by 57.8 percent ($p < 0.01$, column 5), while the protest count estimate is positive but imprecise.¹¹ Protests are the one outcome on which the two measurements disagree as the PPML result in column (6) becomes statistically insignificant.¹²

¹⁰PPML estimates a multiplicative conditional mean, so β is a log-point effect and the implied percentage change is $\exp(\beta) - 1$. For example, the coefficient in Column (4) is $\beta = 0.257$, which corresponds to $\exp(0.257) - 1 = 29.3\%$.

¹¹The per-capita and count coefficients differ in scale for a mechanical reason. A single additional event in a treatment city of 600,000 contributes 0.17 events per 100,000 residents, while the same event in a control city of two million contributes only 0.05. They also rest on slightly different samples, because cities with no recorded events of a given type drop out of the PPML estimation, which is why the observation counts in columns (4)–(6) are smaller. Re-estimating the OLS columns on cities that enter the corresponding PPML estimation leaves the results nearly unchanged.

¹²Repeating columns (1)–(3) using rates with PPML estimation leaves the pooled pattern intact, at 0.327 for unrest and 0.532 for strikes, both significant, with an imprecise protest estimate (Appendix Table A11).

Figure 3: Per Capita Unrest Rates (2007–2019)



Notes: Event study coefficients and 95% confidence intervals for per capita unrest rates (events per 100,000 registered population). Panel (a) reports overall labor unrest, panel (b) reports strikes, and panel (c) reports protests. Sample, specification, and reference year as in Figure 2.

The increase in registered population following the reform dampens the estimated effect when unrest is measured as a rate per registered resident. Because the denominator grows alongside registered population, expressing unrest relative to the contemporaneous population absorbs part of the increase that would be visible relative to a fixed pre-reform population. Columns (7)–(9) illustrate the magnitude of this attenuation. When the denominator is instead fixed at 2010 registered population, the estimated effects are 0.141 for overall unrest, 0.055 for strikes, and 0.084 for protests, each exceeding the corresponding estimates based on the time-varying denominator in columns (1)–(3). The difference reflects the portion of the increase that is mechanically absorbed by the expanding denominator. Thus, the time-varying rate estimates provide a more conservative measure of the increase than the corresponding fixed-baseline rates.

Figure 4 extends the event study to all five city-size groups, using megacities as the common reference category. Size 5 cities (below 0.5 million residents, where the reform eliminated the Hukou requirements entirely) stand apart in overall unrest, with differentials of 0.357, 0.385, and 0.369 events per 100,000 residents in 2014–2016, compared with estimates near zero for the other size groups. More broadly, the post-reform effects follow a clear ordering across the four non-reference groups, with event-study coefficients averaging -0.063 , -0.044 , 0.005 , and 0.198 over 2014–2019 for size 2 through size 5, respectively. Strikes exhibit the same ordering and a sharper gradient. The ordering holds in each post-reform year, and the corresponding pooled estimates are 0.002, 0.029, 0.059, and 0.100.¹³

¹³In Appendix Figure A4, we re-estimate the per capita event study by size category using PPML. Size 5 cities again show the largest effects, with estimates of 1.15 and 0.69 in 2014 and 2015, respectively, both statistically significant, while size 2 cities turn negative from 2015 onward without becoming statistically distinguishable from zero.

Table 1: DiD Estimates for Unrest Outcomes (2007–2019)

	Per capita rates			Event counts			Scaled counts (2010 pop.)		
	Unrest (1)	Strike (2)	Protest (3)	Unrest (4)	Strike (5)	Protest (6)	Unrest (7)	Strike (8)	Protest (9)
Treat $\times$ Post	0.106*** (0.033)	0.043*** (0.010)	0.061** (0.028)	0.257** (0.104)	0.456*** (0.126)	0.262 (0.186)	0.141*** (0.039)	0.055*** (0.010)	0.084** (0.033)
Method	OLS	OLS	OLS	PPML	PPML	PPML	OLS	OLS	OLS
Control post mean	0.383	0.070	0.310	8.127	1.485	6.642	0.465	0.084	0.377
Observations	3,510	3,510	3,510	3,180	3,032	2,501	3,484	3,484	3,484
R ² / Pseudo R ²	0.63	0.47	0.61	0.67	0.50	0.64	0.66	0.49	0.64

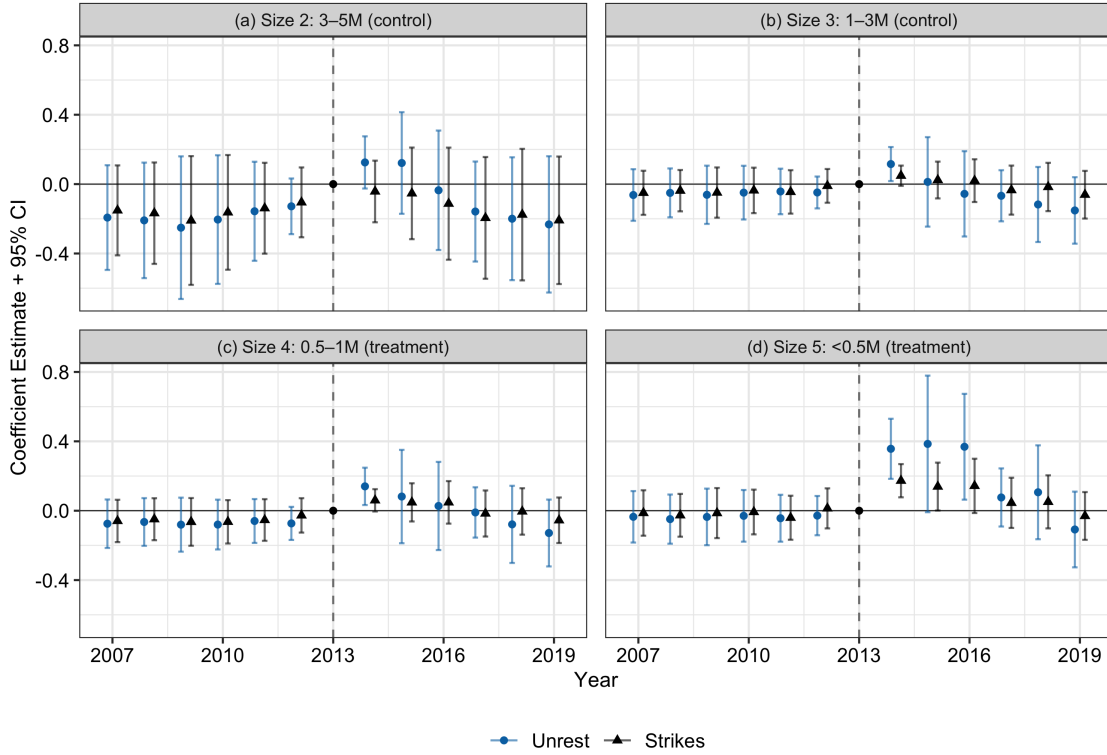
Notes: Columns (1)–(3) report per-capita unrest rates (events per 100,000 contemporaneous registered residents), estimated by OLS. Columns (4)–(6) report PPML estimates on raw event counts. Post = 1 after 2013. The sample excludes size-1 megacities. All models include city and province-by-year fixed effects, with standard errors clustered by city. Columns (7)–(9) repeat the rate specification holding the denominator at 2010 registered population. Control post mean is the control-group mean over 2014–2019, the denominator against which the proportional effects in the text are computed. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The fact that the estimated effects vary systematically with the extent of the reform provides evidence beyond the discontinuity at the 1 million population threshold. The reform did not simply divide cities into treated and untreated groups: its intensity varied systematically with city size, and the estimated effects increase with the extent of the reform. Thus, to explain the results without a treatment effect, a confounding factor would need to vary with city size in a way that generates a similar dose-response pattern across the four treatment groups, rather than merely producing a break between cities above and below the 1 million threshold. Such a pattern is possible if other determinants of unrest also vary systematically with city size, so the heterogeneity analysis does not by itself rule out confounding. Section 3.4 therefore examines the possible confounding factors directly.

In contrast, [Lai and Qiu \(2025\)](#) use a difference-in-discontinuity design around the 3 million population threshold and report a 42 percent reduction in labor unrest. The two designs isolate different margins of the same reform: their cutoff contrasts cities receiving minimal barrier reduction (1–3 million) with those receiving essentially none (3–5 million), whereas ours contrasts the most extensively treated cities with partially treated cities. Applied to our sample, their cutoff yields -0.044 ($p = 0.40$), based on only 11 control cities, with standard errors roughly 60 percent larger than in our baseline (Appendix Table A13); the corresponding event study also shows no break in 2013 (Appendix Figure A2). This null is consistent with our size gradient: a 3-million cutoff places both heavily and barely treated cities on the same side of the comparison, averaging away the treatment-intensity gradient shown in Figure 4.

The remaining differences reflect identification, treatment definition, and outcome measurement. Their estimator absorbs the relationship between city size and unrest with a polynomial in the running variable, making identification dependent on that functional form. Our specification

Figure 4: Per Capita Unrest and Strikes by City Size Category (2007–2019)



Notes: Event study coefficients and 95% confidence intervals per 100,000 registered population, using megacities (size 1, population above 5 million) as the reference group. Panel (a) shows size 2 cities (3–5 million), panel (b) shows size 3 cities (1–3 million), panel (c) shows size 4 cities (0.5–1 million), and panel (d) shows size 5 cities (below 0.5 million). Fixed effects, clustering, and reference year as in Figure 2.

instead compares within cities using city fixed effects.¹⁴ Their treatment indicator also pools 250 reform cities spanning both our treatment and control groups, precluding the treatment-intensity gradient that our five-group specification identifies. Finally, they normalize unrest by prime-age physical population, which their estimates show did not respond to the reform, whereas we normalize by registered population, the margin that did respond. This denominator difference does not account for the divergence: fixing our denominator at its 2010 level produces larger, not smaller, estimated effects (columns (7)–(9) of Table 1). Section 3.3 turns to the sectoral composition of the increase.

3.3 Sectoral Heterogeneity

If the increase in labor unrest operated through migrant labor markets, it should be concentrated in sectors where migrant workers are most prevalent. The CLB records both the industrial sector

¹⁴Without the polynomial control, reform and non-reform cities in [Lai and Qiu \(2025\)](#) exhibit significant pre-existing differences in unrest growth, population growth, and migrant shares (their Table 2). Our panel provides six estimable pre-reform coefficients, compared with two in their design.

and participant count for each incident, allowing this prediction to be tested directly. We aggregate the raw unrest categories into *Manufacturing* (manufacturing, heavy industry, and mining), *Construction, Transportation* (transport, storage, logistics, and postal), and *Services* (education, the public sector, wholesale and retail trade, hotels and catering, medical services, sanitation and other service activities). Services act as the comparison category, allowing us to distinguish a sector-specific migrant-labor mechanism from a general increase in unrest associated with smaller cities. A city-size shock alone need not generate the same sectoral pattern.

The CLB records participants in four bins (1–100, 101–1,000, 1,001–10,000, and 10,000+). We follow this coding scheme and classify incidents with 1–100 participants as small-scale and those with more than 100 participants as large-scale. Figure 5 reports per capita event-study estimates for the eight categories formed by crossing the four sectors with the two scales, using fixed effects and sample of the baseline specification.

The per capita increase in unrest is concentrated in manufacturing, construction, and transportation, the three sectors that employ most migrants.¹⁵ Worker unrest events in small-scale manufacturing exhibit a sharp post-reform increase in the event-study estimates, with coefficients rising from near zero in 2013 to around 0.03–0.07 events per 100,000 population in 2015–2016. In addition, worker unrest events in large-scale manufacturing display a more persistent positive differential, between 0.02 and 0.04, throughout the post-reform period.¹⁶ Construction unrest incidents also rose, with small-scale construction events rising to approximately 0.09 per 100,000 by 2016.¹⁷

Appendix Table A8 reports the pooled estimates for all eight categories. The three migrant sectors are positive and significant at both incident sizes, apart from large-scale construction, where disputes of 100 or more participants are rare in cities this small. Neither service category moves, which is what a migrant channel likely predicts and what a shock loading on city size would not produce. Transportation shows a smaller level differential than manufacturing or construction, but transportation unrest is also far rarer per resident, so the level understates the response. Measured against the control group’s post-reform rate in the same cell, small-scale manufacturing rises 23 percent, small-scale construction 26 percent, and small-scale transportation 85 percent, while small-scale services rises 2 percent. Transportation is also where a competing explanation is most obvious, and Section 3.4 takes it up directly.

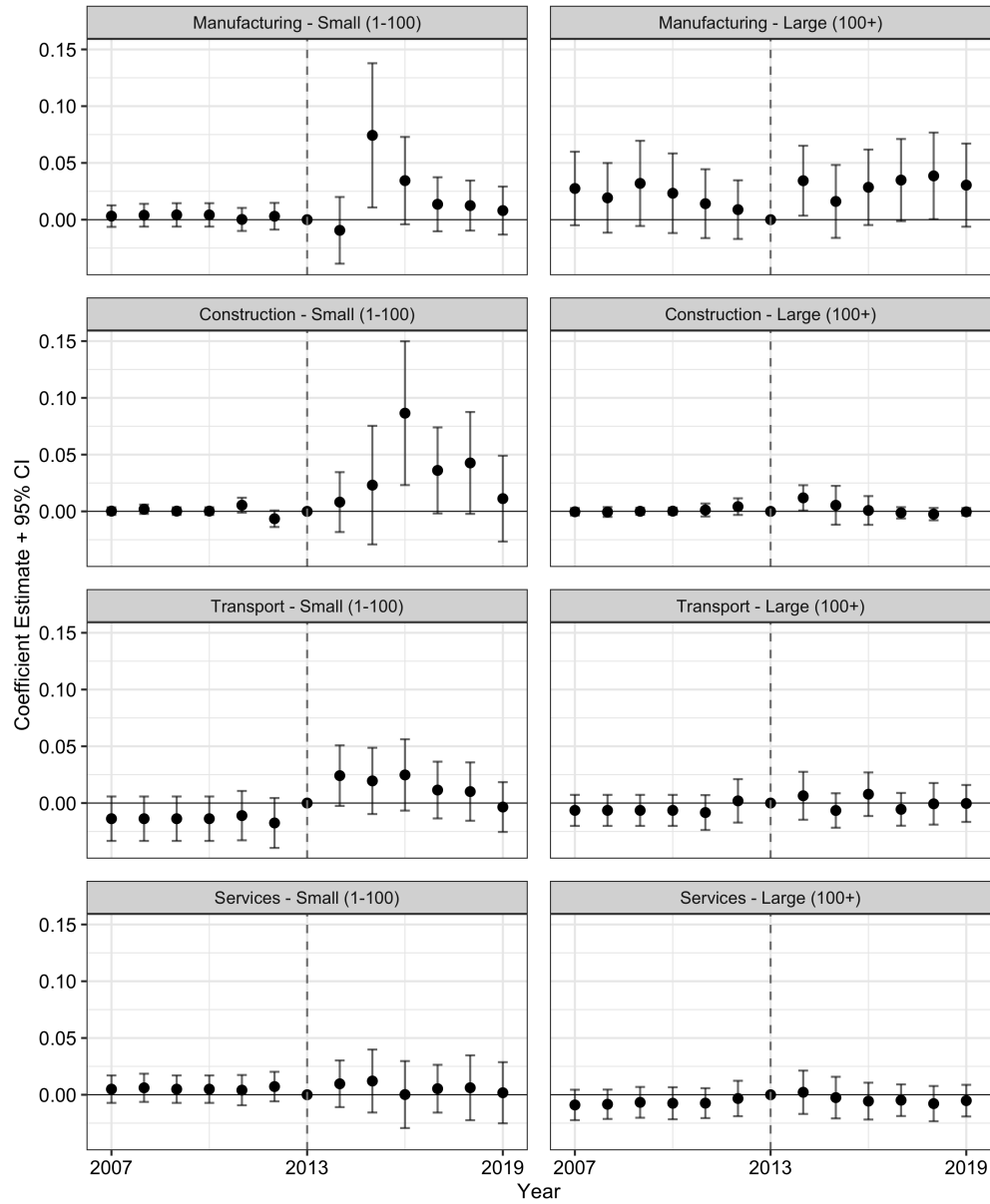
The rates in Figure 5 measure how much unrest a resident of a treatment city encountered in

¹⁵The National Bureau of Statistics Migrant Worker Survey, covering 31 provinces and approximately 1.5 million observations, reports that manufacturing and construction together provide 52–55 percent of employment for migrant workers nationally, making them the two largest entry sectors (National Bureau of Statistics of China, 2015). Transport, storage, and postal services account for an additional share.

¹⁶The Manufacturing–Large panel in Figure 5 is positive in every pre-reform year, with coefficients from 0.009 to 0.032 events per 100,000 residents. None is distinguishable from zero at the 5 percent level, and the joint test of the six does not reject ($p = 0.182$, Appendix Table A4). The Manufacturing–Small panel has a cleaner flat pre-trend and provides the sharper event-study evidence.

¹⁷Appendix Table A9 shows that construction events are sparse before electronic CLB coverage begins in 2011. Small-scale construction appears only once in 2007–2010, in treatment cities in 2008, so the earliest construction coefficients are estimated from very limited raw variation.

Figure 5: Per Capita Unrest by Sector and Incident Scale (2007–2019)



Notes: Event study coefficients and 95% confidence intervals for per capita unrest (events per 100,000 registered population) by sector and incident scale. Rows show sectors and columns show incident scale. Manufacturing covers manufacturing, heavy industry, and mining. Construction covers construction. Transport covers transport, storage, logistics, and postal services. Services covers education, the public sector, and the remaining service industries, principally wholesale and retail trade, hotels and catering, medical, and sanitation. Small = 1–100 participants. Large = 100+. Sample, specification, and reference year as in Figure 2.

each sector as they are divided by total registered population. Sector composition matters if treatment cities shift toward the relevant sector after the reform: sector incidents per resident could rise even if incidents per sector worker did not. Appendix Figure A5 and Table A10 check this possibility by rescaling manufacturing and construction incidents by secondary-sector employment, and transport and service incidents by tertiary-sector employment. We find that manufacturing at both incident sizes, small-scale construction and small-scale transportation remain positive and significant, and both service categories remain indistinguishable from zero. The one cell that moves is large-scale transportation, marginal per resident and insignificant per worker. Measured the same way, the employment denominator gives 46 percent for small-scale manufacturing, 37 percent for construction, and 51 percent for transportation, against 23, 26, and 85 percent per resident. Transportation is the largest either way, although rescaling compresses the spread and reverses the order of the other two. Whichever denominator is used, services sector does not respond.

The sectoral pattern invites a question about who took part in these unrest events. The CLB records the sector, scale, and stated grievance of each incident but not the Hukou status of the people involved. As a result, we cannot provide a definite answer to this question using the data we have.

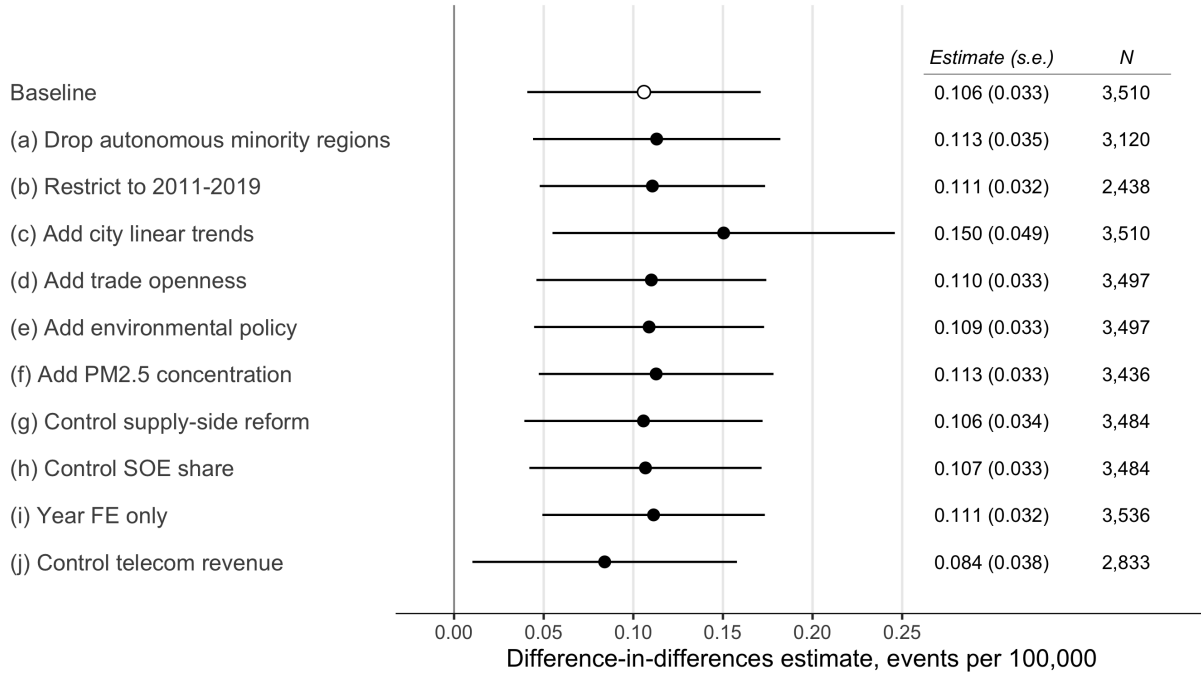
3.4 Robustness Analysis

The baseline per capita finding could reflect sample selection, concurrent shocks, violations of parallel trends, the choice of comparison group, or differential reporting rather than the Hukou reform itself. We consider these possibilities in turn. The reporting concern carries the most weight, since CLB coverage expanded over the same years as the reform, and we address it both through the severity profile of our own estimates and through a separately collected event series.

Alternative specifications. We report per capita unrest estimates under ten alternative specifications in Figure 6. Specification (a) excludes autonomous minority regions (Tibet, Xinjiang, Guangxi, Inner Mongolia, and Ningxia), where unique political and societal factors could drive labor unrest independent of Hukou policy changes. Specification (b) restricts the sample to 2011–2019, the period with consistent electronic CLB coverage, dropping the pre-2011 years reconstructed from annual reports. The estimates remain virtually identical to the baseline in both cases. Specification (c) adds city-specific linear time trends, addressing gradual differential city trajectories that may not be fully captured by pre-trend tests. These trends also help account for slowly evolving city-level labor-market conditions, including differential minimum wage trajectories that can vary across cities within a province. The estimated effect is larger than the baseline, 0.150 against 0.106, because the trends absorb slow-moving differences in city trajectories that otherwise work against the post-reform break.

Specifications (d) through (h) address specific concurrent shocks that could confound the baseline estimate. Specification (d) controls for city-level trade openness, since China’s export slowdown after 2014 differentially affected manufacturing-heavy smaller cities and could indepen-

Figure 6: DiD Estimates Under Alternative Specifications (2007–2019)



Notes: Each row reports the pooled DiD estimate ($\text{Treat} \times \text{Post}$) for per capita overall unrest (events per 100,000 registered population) with its 95 percent confidence interval, alongside the estimate, its standard error, and the number of city-year observations. Unless a row states otherwise, sample and specification as in Figure 2. Specification (b) keeps the years of consistent electronic CLB coverage. Specification (g) interacts each city’s 2010 secondary employment share with an indicator for years from 2015 onward. Specification (h) interacts the 2010 urban unit employment share with the post-reform indicator. Specification (i) replaces province-by-year fixed effects with year fixed effects only. Specification (j) controls for log per-capita post and telecommunications revenue.

dently generate labor unrest (Campante et al., 2023). Specifications (e) and (f) address environmental regulation. Specification (e) adds interaction terms between year fixed effects and an indicator for the “2+26” cities subject to intensified environmental monitoring and factory shutdowns post-2014,¹⁸ and specification (f) adds city-level PM2.5 concentrations (Khanna et al., 2025; Li and Meng, 2023), since pollution-related protests are a documented form of collective action in China. Specification (g) controls for the supply-side structural reform campaign that began in 2015, which shut excess capacity in coal, steel, and cement and could have generated layoff-driven unrest in manufacturing-heavy cities, by interacting each city’s 2010 secondary employment share with an indicator for years from 2015 onward.

Specification (h) addresses the anti-corruption campaign that intensified after 2012, which disrupted patronage networks and could provoke protest in cities with larger state-owned enterprise presence, by interacting the 2010 urban unit employment share with the post-reform indicator. Specification (i) replaces province-by-year fixed effects with year fixed effects only. The estimate

¹⁸The “2+26 Cities” refers to a regional air-quality management strategy that includes Beijing, Tianjin, plus 26 nearby prefecture-level cities in northern China (Shu et al., 2022).

remains virtually unchanged (0.111 versus 0.106 in the baseline), confirming that the result does not depend on the more demanding fixed effect structure. Specification (j) controls for (log) per-capita post and telecommunications revenue, a proxy for communication-technology diffusion. Every interval in Figure 6 excludes zero, and eight of the ten estimates sit within 0.01 of the baseline. The two that move are specification (c), where city-specific trends raise the estimate to 0.150, and specification (j), which we take up under data validity below.

Inference. Appendix Table A4 reports joint F -tests for every event study in the paper, eighteen in all, and none rejects. They cover the three OLS outcomes and their count PPML versions, the eight sector-by-scale panels behind Figure 5, the registered-population and sectoral-employment event studies that anchor the mechanism analysis, and the specification adding city-specific linear trends. No p -value falls below 0.104, and the lowest belongs to small-scale construction, the cell the paper already flags as sparse before electronic coverage begins in 2011. Appendix Figure A1 presents a sensitivity analysis following [Rambachan and Roth \(2023\)](#). The confidence interval for the pooled post-treatment effect excludes zero when post-treatment violations of parallel trends are permitted to be up to half as large as the maximum pre-reform violation ($\bar{M} = 0.5$), and includes zero at $\bar{M} = 1.0$.

Alternative control groups. We replace the control cities with megacities (population above 5 million), a comparison group that received no barrier reduction at all. Appendix Figure A6 shows the event-study dynamics and Appendix Table A12 reports the pooled DiD estimates using contemporaneous registered population. The results are consistent with the baseline. Strikes are positive and statistically significant (0.068, $p < 0.01$). Overall unrest is positive but less precise (0.107), because the megacity shocks set out in Section 2.3 enter the counterfactual once megacities serve as controls. The registered-population response against this benchmark is similar in magnitude to the baseline first stage but imprecisely estimated.¹⁹ We read this comparison as the same mechanism measured against a weaker benchmark, since twelve cities carry less power than the baseline contrast, and the consistency of the strike results indicates that the finding is not an artifact of the control group.

We can also narrow the comparison instead of widening it, using only cities of 1–3 million as controls for our treatment cities. This restriction drops the largest control cities (3–5 million) and asks whether our main results hold against a more homogeneous comparison group. Appendix Table A13 shows that they do. The estimated effect increases with reform intensity, 0.070 per 100,000 for cities with population 0.5–1 million ($p < 0.05$) and 0.233 for cities below 0.5 million ($p < 0.01$). Appendix Figure A3 shows broadly flat pre-trends and a post-reform increase for both treatment groups, with larger effects for the smallest cities. The ordering is what the reform predicts.

¹⁹The registered-population differential against megacities is 0.055 log points (s.e. 0.077), close to the baseline estimate of 0.057.

Data validity. As discussed in Section 2, CLB coverage expanded substantially over the sample period, coinciding with the years of the reform. If detection improved faster in treatment cities, part of the estimated post-reform increase could reflect improved observation rather than more unrest. This concern has two distinct variants. First, reporting in conflict-event data depends on communication infrastructure (Weidmann, 2016), thus the diffusion of smartphones into smaller cities could mechanically increase the number of recorded incidents. Second, greater access to social media could increase actual protest participation by lowering coordination costs (Enikolopov et al., 2020).

The severity profile of the estimated effects speaks most directly to the first, mechanical channel. Reporting rates increase sharply with event severity (Weidmann, 2016), implying that improved detection should primarily affect small, low-salience incidents that were previously difficult to observe. Instead, the estimated effects remain positive and statistically significant for incidents involving more than 100 participants, both in the sector-by-scale decomposition in Section 3.3 and for the headline outcomes: 0.021 (s.e. 0.008) for overall unrest and 0.015 (s.e. 0.006) for strikes (Appendix Table A6). Events involving hundreds of workers are unlikely to be the marginal incidents newly brought into the data by improved smartphone-based reporting. The large-event results therefore make a pure detection explanation less plausible.

Three further features of the data provide additional evidence against a pure reporting explanation. First, the CLB is a nongovernmental organization based in Hong Kong that compiles incidents from social media, overseas Chinese media, and labor activists (Qin et al., 2024; Elfstrom and Kuruvilla, 2014), rather than relying on local government reporting. Second, the CMDS is an independent government survey unaffected by changes in CLB coverage; among surveyed non-local-Hukou migrants, it shows wage declines in the same sectors in which unrest increased, subject to the converter-exit limitation discussed in Section 4.3. Third, controlling for (log) per-capita post and telecommunications revenue, our closest available proxy for communication-technology diffusion (Campante et al., 2023), reduces the estimate from 0.106 to 0.084 while leaving it statistically significant at the 5 percent level (specification (j) of Figure 6). The attenuation suggests that communication technology may account for part of the estimated increase, but the remaining effect is inconsistent with technology-driven reporting or participation being the sole explanation. Because this control captures communication technology regardless of whether it affects recorded events or actual participation, while province-by-year fixed effects absorb changes common to cities within a province, the result addresses both variants of the concern. We next turn to an outcome collected independently of the CLB.

Separate Radio Free Asia (RFA) reporting placebo. A detection artifact should raise every kind of recorded collective action in smaller cities, not labor unrest alone. A second event series that shares the observability environment but not the labor-market mechanism therefore discriminates between the two explanations, and the protest data in Qin et al. (2024) provide one. These incidents are collected by RFA from its correspondents and from Chinese and international news

sources, and they are distinct from the China Labour Bulletin (CLB) records behind our labor-unrest outcomes. We examine four outcomes, ordered by their distance from labor disputes, namely all RFA protests, non-labor protests, government-related protests (government policy or corruption, police and court, and housing and land), and a narrower political subset (government policy or corruption, police and court, human rights, and ethnic or religious causes). The comparison mirrors the primary OLS specification in every respect, over the 2007–2017 span the RFA data cover.

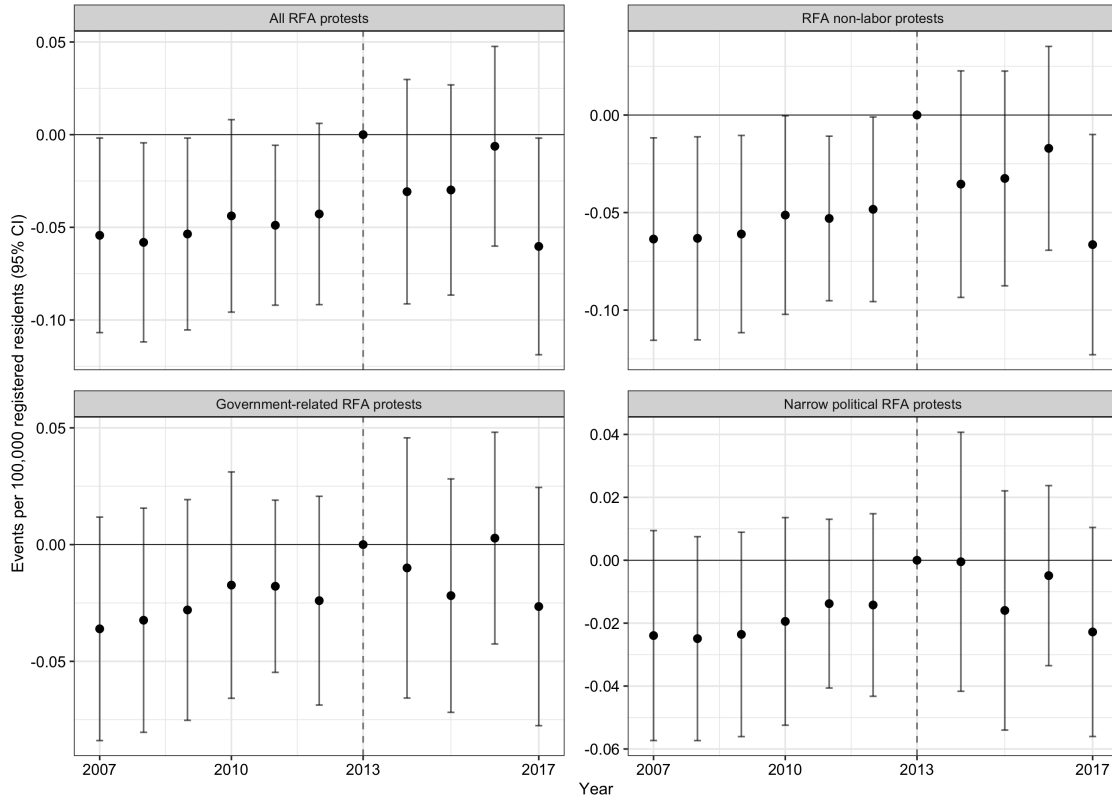
Appendix Figure A7 displays the raw series. Government-related RFA counts rise by 0.178 events per treated city-year against 0.421 among controls, so the raw difference-in-differences is negative (−0.243). In the matched OLS specification none of the four pooled estimates is distinguishable from zero, at 0.011 (s.e. 0.010) for all events, 0.011 (0.010) for non-labor, 0.008 (0.008) for government-related, and 0.006 (0.005) for narrow political events, and using size 3 alone as the adjacent control produces the same conclusion. Figure 7 presents the OLS event studies, where post-reform coefficients are jointly insignificant for all events ($p = 0.122$), government-related events ($p = 0.476$), and narrow political events ($p = 0.294$), with the non-labor path marginal at the 10 percent level ($p = 0.077$). Fifteen of the sixteen post-reform coefficients are negative and the sixteenth is 0.003, so the panels sit at or below their 2013 reference throughout.

Ride-hailing entry. The transportation results could partly reflect the expansion of ride-hailing platforms into lower-tier cities over 2014–2016, which generated driver protests over platform competition and unlicensed vehicles through a channel unrelated to the Hukou reform. We cannot observe city-level platform entry timing, so we take an alternative route and drop the sector entirely. Appendix Table A5 re-estimates the primary per-capita specification after excluding every incident recorded in transportation, storage, logistics, and postal services, including the taxi incidents in the 2007–2010 records. The pooled estimates remain positive and statistically significant for all three outcomes (overall 0.076, s.e. 0.030, strikes 0.016, s.e. 0.006, and protests 0.060, s.e. 0.028). Transportation incidents account for more than a quarter of the overall per-capita differential and more than half of the strike differential, consistent with the sector’s prominence in the heterogeneity analysis, but the headline result does not depend on them.

4 Mechanisms

The concentration of post-reform unrest in manufacturing, construction, and transportation raises a mechanism question. Why these sectors and not others? The population evidence narrows the field. The registered population rose 5–12 percent while the physical population showed no detectable change (Section 3.1), thus the reform’s dominant margin was granting formal status to workers who were already present, with at most a modest inflow of new arrivals. Accounts that require a large physical inflow, including native backlash against newcomers and physical congestion of roads, schools, and hospitals, are hard to square with this evidence. The evidence bears on how many migrants arrived, not on how the reform acted. A worker who gains local registration

Figure 7: RFA Protest Placebo Event Studies, Per Capita OLS Estimates (2007–2017)



Notes: OLS event-study coefficients and 95% confidence intervals for separately collected Radio Free Asia (RFA) events per 100,000 contemporaneous registered residents, estimated with the paper's main specification. Every regression uses city and province-by-year fixed effects, standard errors clustered by city, sizes 4–5 as treatment, and sizes 2–3 as controls. The omitted reference year is 2013. Size-1 megacities are excluded. Raw trends appear in Appendix Figure A7.

and one who arrives already holding it face the same gap between formal entitlement and delivered conditions, so the mechanism below holds on either margin. The margin the reform moved was primarily institutional. Workers living and working in treatment cities acquired formal local residency, and the same cities became easier for new migrants to enter.

Three channels can connect formalization of labor status to labor unrest. The first is a gap between the entitlements enabled by formal status and what employers and local governments actually provide. Hukou registration grants access to local social insurance and public services and establishes a more permanent stake in the destination city, while leaving labor-contract and dispute-resolution rights unchanged: the 2008 Labor Contract Law already extends these protections to employees regardless of registration status. Formalization therefore increased workers' claims on local institutions without changing the jobs they held. The second is congestion in migrant-intensive sectors. When more workers seek employment in sectors with limited job opportunities, competition for existing positions can intensify even without substantial wage adjustment (Harris and Todaro, 1970; Fields, 1975). In this setting, the additional competition need not come from a large physical inflow: newly formalized workers may reallocate across sectors and compete for ex-

isting positions. The third is the decreasing barriers to collective action when residency becomes more secure and permanent. These three mechanisms can reinforce one another: formalization raises expectations, intensifies competition, and strengthens workers' incentives and ability to act collectively, with the effects expected to be most pronounced in manufacturing, construction, and transportation, where informal and piece-rate employment arrangements are prevalent.

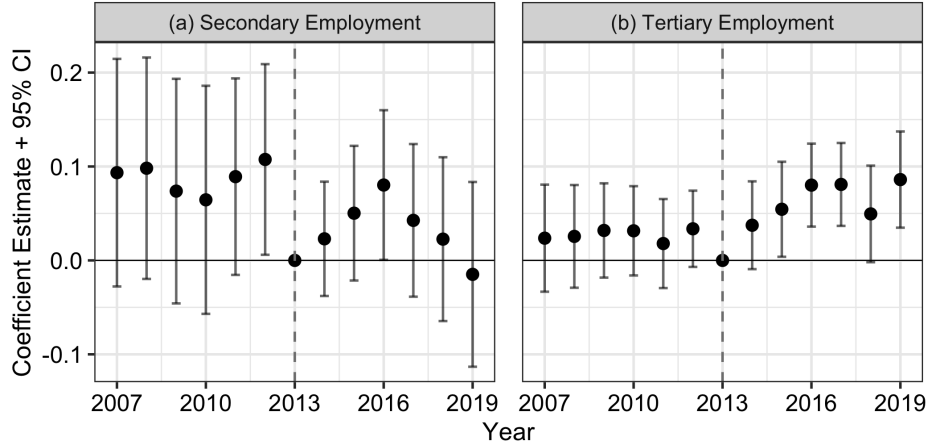
Two descriptive patterns motivate this mechanism. First, social insurance coverage among new migrants fell by 2.1 percentage points in treatment cities after the reform (Appendix Table A14), indicating that formalization did not necessarily translate into improved workplace benefits. Second, wage arrears account for 78 percent of post-reform CLB events. The wage-setting evidence in [Guo et al. \(2025\)](#), which shows that Hukou eligibility enters the worker-employer wage bargaining as a compensating differential within firms, provides a direct link between registration status and employment conditions. A change in registration status can therefore alter workers' expected returns without changing their jobs, creating scope for conflict when promised or expected conditions are not realized.

Two assumptions underpin this argument. First, workers evaluate what they receive against what their formal status entitles them to and act when the resulting gap exceeds the cost of collective action. Formalization moves both terms: it raises the set of entitlements while lowering the cost of acting by providing more secure and permanent residency. Second, employment capacity in migrant-intensive sectors is fixed in the short run. The reform does not immediately expand the stock of manufacturing jobs or the volume of freight and passenger activity over which transport workers compete, limiting the scope for realized earnings to adjust fully to the expansion in entitlements. Under these assumptions, the effect of formalization becomes conditional rather than universal, reconciling our findings with the labor legalization literature. Formalization can reduce unrest when the delivery of benefits and services rises with workers' new entitlements, but increase unrest when entitlements outpace delivery. Because both formal entitlements and the cost of collective action change for workers already residing in the city, the mechanism does not require a physical inflow of migrants.

These three channels generate four testable predictions. First, employment in the secondary sector should not expand differentially in treatment cities, particularly in manufacturing and construction, where short-run capacity is relatively fixed. Second, growth in the tertiary sector should be concentrated in accessible service activities rather than reflecting broad expansion across services. Third, the gap between formal entitlements and realized benefits should be larger in the sectors where unrest increases, and be visible through the sector's pay structure. In manufacturing, relatively rigid wages should shift the adjustment toward job rationing and wage arrears; in transportation, where piece-rate pay is more flexible, the same underlying gap should instead appear as earnings compression. Fourth, the effect should be largest during the initial adjustment period and diminish as workers and employers adapt.

We next examine employment growth, wages, and fiscal capacity against these predictions.

Figure 8: Employment Responses by Sector (2007–2019)



Notes: Event study coefficients and 95% confidence intervals for log employment. Panel (a) reports secondary-sector employment and panel (b) reports tertiary-sector employment. Joint F -tests cannot reject zero pre-reform coefficients in either panel (Appendix Table A4). Sample, specification, and reference year as in Figure 2.

4.1 Sectoral Employment Responses

Figure 8 presents event-study estimates for (log) employment in the secondary sector (including manufacturing) in panel (a) and the tertiary sector (services) in panel (b), comparing treatment and control cities. The two sectors show sharply different patterns. In panel (a), secondary-sector employment does not show a clear post-reform break. The pre-reform coefficients are positive and somewhat uneven, although the joint test does not reject the null of parallel pre-trends ($p = 0.348$, Appendix Table A4). The post-reform coefficients remain close to zero with wide confidence intervals, and the pooled DiD estimate is -0.041 (Appendix Table A15), statistically indistinguishable from zero. Thus, the reform did not produce additional differential growth in secondary-sector employment. The absence of an employment response is consistent with limited short-run capacity to absorb the newly formalized workforce in manufacturing and related activities.

Tertiary employment responds differently. Panel (b) shows a clear post-2013 increase, with coefficients rising to 0.05 – 0.09 by 2016–2019. The pooled DiD estimate of 0.041 ($p < 0.10$, Table A15) corresponds to approximately four percent higher service employment in treatment cities after the Hukou reform. Thus, employment expanded in services but not in the secondary sector. The reform therefore coincided with an expansion of employment in services, while secondary-sector employment remained essentially unchanged. Section 4.2 traces which service subsectors account for that growth.

4.2 Labor Absorption in the Service Sector

The aggregate expansion in tertiary employment discussed in the previous section leaves open which service activities account for the increase. This distinction matters for the congestion mech-

anism because sectors differ in how readily they can absorb additional workers. In activities with relatively low entry barriers, such as transportation and logistics, employment can expand more readily, shifting the adjustment toward earnings rather than employment. In subsectors that require greater physical capital or formal qualifications, employment may be less responsive, leaving congestion to appear as rationing of available positions. Workers queue for the limited positions they can actually fill. Appendix Figure A8 decomposes tertiary employment into six subsectors to identify where the expansion occurred.

Among the six subsectors, transportation and communication is the only one with a statistically significant expansion (Column 4, Appendix Table A15). The pooled DiD estimate of 0.101 ($p < 0.10$) implies approximately 11 percent higher transportation employment in treatment cities after the reform, or roughly 430 additional workers per city. Treatment-city transportation employment averaged about 3,250 before the reform ($\exp(8.09)$), while transportation employment in control cities increased by approximately 30 percent over the same period, implying a counterfactual post-reform level of roughly 4,200 workers. The event study shows a clear break in 2014: coefficients move from approximately -0.05 before the reform to 0.04 – 0.10 afterward, with no evidence of a differential pre-trend.

Wholesale and retail trade follows a similar pattern but is not statistically significant: the pooled estimate is 0.070, with event-study coefficients of 0.07 – 0.13 before moderating in later years. The remaining four subsectors show no statistically significant differential expansion (pooled estimates in parenthesis): healthcare (0.083), social services (0.048), scientific and technical services (0.037), and finance (0.018).²⁰

The transportation sector presents an apparent paradox. Employment expanded by approximately 11 percent, yet per capita unrest in the sector also increased. One possible explanation is that transportation services absorbed additional workers without generating an increase in earnings. Urban transportation in China relies heavily on piece-rate compensation, with drivers paid per trip or delivery. When more workers compete for a given volume of freight or passengers, employment can expand while earnings per worker decline. The composition of CLB incidents is consistent with this possibility. Among the 499 transportation unrest events recorded in treatment cities during 2014–2019, 31 percent cite earnings-related grievances, including fare disputes, wage arrears, and fee complaints, while 46 percent cite competitive pressure, including ride-hailing platforms and unlicensed vehicles; 68 percent mention at least one of the two categories.²¹ Thus, the sector expanded employment while possibly compressing the earnings associated with each position, and the composition of grievances indicates that these pressures were contested rather than fully absorbed by workers. The individual-level wage evidence in the next subsection provides a direct test of this earnings-compression mechanism.

²⁰Scientific and technical services exhibits some positive pre-reform drift in the event study, therefore its post-reform trajectory should be interpreted cautiously.

²¹We classify incident descriptions using pre-specified keywords. Earnings terms cover fare, fee, deposit, unpaid wage, salary, income, and commission disputes. Competition terms cover ride-hailing platforms, unlicensed vehicles, illegal taxis, and oversupply. Categories overlap because some events mention both earnings grievances and competition.

These patterns reflect differential adjustment speeds across service sectors. Transportation, retail, and logistics are relatively low-capital services with lower credential requirements, allowing employment to expand quickly when labor supply increases. These sectors draw from the same pool of workers as manufacturing, which, together with construction, accounts for 52–55 percent of migrant employment nationally ([National Bureau of Statistics of China, 2015](#)). By contrast, higher-skill services such as finance and healthcare require greater physical capital investment, specialized qualifications, or regulatory approval, limiting the speed at which they can expand. Even when these sectors eventually grow, the workers they attract are less likely to have been competing for manufacturing jobs in the first place.

Accessible employment therefore expanded in a narrow set of service subsectors, adding roughly 3,100 tertiary-sector positions per treatment city relative to a counterfactual base of about 72,000. Workers who obtained local registration and sought to translate it into better employment opportunities therefore faced a manufacturing sector with no differential expansion, while the accessible service margin that did expand absorbed workers under conditions consistent with earnings compression. The adjustment thus operated along two margins: quantities, through limited expansion of employment opportunities in migrant-intensive sectors, and prices, through lower earnings in the service margin that did expand. Both patterns are consistent with the congestion mechanism.

4.3 Migrant Wages and Sectoral Reallocation

The employment dynamics described above raise the question of whether wage pressure and worker reallocation are observable directly in individual-level data. To address this question, we use the China Migrants Dynamic Survey (CMDS) and examine both aggregate wages and sector-specific patterns using Equation (2).²² The survey requires respondents to have lived in the destination for at least one month while holding Hukou registered outside the local county or city. Migrants who obtain local Hukou therefore exit the sampling universe, and the survey does not follow individuals after conversion. Conversion status is consequently unobserved, therefore the estimates apply to surveyed non-local-Hukou migrants and may partly reflect selection from exit. We follow the specification in [An et al. \(2024\)](#), who use the same survey to study the wage effects of the Hukou reform:

$$\ln w_{ict} = \alpha + \beta \text{Treat}_c \times \text{Post}_t + X'_{ict} \gamma + \mu_c + \delta_{p(c),t} + \phi_{o(i),t} + \varepsilon_{ict} \quad (2)$$

where w_{ict} is the monthly nominal wage of migrant i in city c at time t . X_{ict} is a set of control variables and includes gender, age, age squared, education, marital status, and Hukou type. μ_c are city fixed effects, $\delta_{p(c),t}$ are province-by-year fixed effects, and $\phi_{o(i),t}$ are occupation-by-year fixed effects. Standard errors are clustered at the city level. We first estimate Eq. (2) pooling all sectors, then separately for manufacturing, construction, transportation, and other services to

²²The CMDS covers approximately 485,000 migrant workers across our sample cities from 2011 to 2017, providing only three pre-reform years (2011–2013), which limits the scope for diagnosing pre-trends in wage outcomes. Appendix Table A16 reports summary statistics.

identify where wage pressure was concentrated. Panel B replaces the dependent variable with a sector indicator and uses the full sample of migrants with non-missing industry codes to test whether the reform shifted which sectors migrants work in.

Table 2 presents the results. Province-by-year fixed effects absorb province-level price movements in each year, so identification comes from within-province differences across city sizes, net of common price changes. Among migrants who remain in the CMDS sampling frame, Panel A, column (1) shows that wages declined by 3.7 percent in treatment cities relative to control cities ($p < 0.01$). Columns (2)–(5) decompose the estimate by sector. Wages fell significantly in construction (−4.1 percent, $p < 0.01$), transportation (−3.8 percent, $p < 0.05$), and other services (−4.1 percent, $p < 0.01$), precisely the sectors in which per capita unrest increased. Manufacturing wages declined by 1.7 percent, but the estimate is not statistically significant. Appendix Table A17 shows that this result is stable across alternative specifications. These sectoral patterns are consistent with [An et al. \(2024\)](#), who use the same survey and find average migrant wage declines of 2.6–7.9 percent in non-megacities, with larger effects among less-educated workers in the smallest cities.

The absence of a statistically detectable wage decrease in manufacturing is also consistent with institutional constraints on downward wage adjustment. Minimum-wage requirements and firm-level wage-setting arrangements may limit the scope for nominal wage cuts, shifting adjustment toward other margins. Consistent with this interpretation, CLB incident descriptions in manufacturing predominantly cite wage arrears and unpaid overtime rather than direct wage reductions. The contrast with transportation is informative. Where compensation is flexible and closely tied to output, congestion can manifest as lower realized earnings; where wages are more rigid, similar labor-market pressure may instead appear through arrears, unpaid hours, or reduced access to employment. The wage evidence thus accords with the two-margin adjustment mechanism described above: congestion can operate through prices when earnings are flexible and through non-wage employment conditions when they are not.

Panel B asks whether the reform shifted which sectors migrants work in. The only significant reallocation is into other services (+1.7 percentage points, $p < 0.01$, on a pre-reform base of 44.1 percent). Manufacturing, construction, and transportation shares are unchanged (Appendix Table A17 confirms these results across alternative fixed effect structures). Migrants entered the most accessible service occupations, including delivery, catering, and resident services, where wages fell as a result. The cross-sector reallocation margin parallels [Guo et al. \(2025\)](#), who document that changes in Hukou-quota access reallocate migrants across sectors and trace the wage consequences within employers. Their evidence captures the wage margin of that reallocation, while ours captures the unrest margin.

Appendix Table A18 decomposes the wage effect by migrant tenure. The wage decreases concentrate among longer-tenure migrants, incumbents already present when the reform took effect, rather than among recent arrivals, and the interaction between treatment and recent arrival status is insignificant in all four sectors. The reform did not protect incumbents from wage pressure,

Table 2: Migrant Wages and Sector Choice (2011–2017)

	All (1)	Manufacturing (2)	Construction (3)	Transportation (4)	Other services (5)
<i>Panel A. Log wage</i>					
Treat $\times$ Post	-0.037*** (0.009)	-0.017 (0.012)	-0.041*** (0.015)	-0.038** (0.015)	-0.041*** (0.012)
Pre-reform mean	7.885	7.876	8.028	7.777	7.892
Observations	424,723	99,093	36,883	59,106	229,622
R ²	0.26	0.36	0.27	0.26	0.24
<i>Panel B. Sector choice (= 1 if in sector)</i>					
Treat $\times$ Post		0.003 (0.004)	-0.004 (0.002)	-0.001 (0.002)	0.017*** (0.005)
Pre-reform mean		0.246	0.082	0.121	0.441
Observations		485,496	485,496	485,496	485,496
R ²		0.61	0.62	0.64	0.53

Notes: Panel A estimates Eq. (2). Wages are in nominal values. Column (1) pools all sectors. Columns (2)–(5) restrict to migrants in the indicated sector. Panel B replaces the dependent variable with a sector indicator and uses the full sample. All specifications include individual controls (gender, age, age squared, education, marital status, Hukou type), city, province-by-year, and occupation-by-year fixed effects. Standard errors clustered at the city level. Data from the CMDS (2011–2017). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

consistent with a broad congestion effect rather than a compositional shift driven exclusively by new entrants. How far the decomposition can be pushed is limited by the sampling frame, since long-tenure migrants are precisely those most likely to obtain local Hukou and exit the survey, so the split cannot separate wage pressure on incumbents from the selective exit of successful converters. We read these patterns only as suggestive.

Appendix Table A19 reports city-level average wages and total fiscal outcomes. Average city wages show no differential change (-0.006 , s.e. 0.014 , column 1), which is consistent with a composition difference between city averages and migrant-specific wages.²³ Log total revenue and log total expenditure likewise show no differential change (-0.020 , s.e. 0.039 , column 2, and 0.006 , s.e. 0.037 , column 3). Because the registered population grew faster in treatment cities, per-capita fiscal ratios decline mechanically even as total resources hold steady.

Beyond the grievances themselves, Hukou formalization may also have strengthened migrants' willingness to act on them. A worker with permanent local residency has a longer horizon in the city, a lower cost of remaining after a dispute, and less to lose from employer retaliation than a worker whose presence is administratively provisional.

The mechanism evidence provides three stylized facts. First, secondary employment did not

²³Native wages showed no corresponding decline and were slightly positive (An et al., 2024), so in a city-level average the negative migrant effect and the flat native effect largely cancel.

expand differentially in treatment cities. Second, the growth in tertiary employment concentrated in transportation, a low entry-barrier sector. Third, wages among surveyed non-local-Hukou migrants fell in sectors where unrest increased. These patterns are consistent with a formalization gap amplified by congestion in the sectors where formalized workers were already employed, though the CMDS sampling-frame limitation prevents a definitive mechanism ranking. Distinguishing the formalization gap, sectoral congestion, and the stake effect would require worker-firm data on Hukou conversion, insurance enrollment, grievance filing, and payment practices that are not available at the city-year level.

5 Conclusion

We find that per capita labor unrest increased by 0.106 events per 100,000 registered residents in cities that received the most extensive barrier reduction under China's 2014 Hukou reform. This corresponds to an increase of roughly 30 percent in recorded events relative to the control-group post-reform level. The differential with control cities peaked in 2016 and was concentrated in manufacturing, construction, and transportation, where informal and piece-rate employment arrangements are prevalent. The effects also increase systematically with reform intensity: cities receiving deeper barrier reductions experienced larger increases in per capita unrest, with the ordering of strike effects closely matching the ordering of reform intensity. The effects attenuated by 2019, consistent with adjustment as workers and employers adapted to the new institutional environment. A second protest series compiled independently from the same online environment shows no comparable increase, making a pure reporting explanation less consistent with the evidence.

Taken together, the results are consistent with a formalization gap. The reform expanded workers' formal-status entitlements without producing differential growth in secondary-sector employment. The expansion of tertiary employment was concentrated in a narrow, low-barrier subsector, while wages declined among surveyed non-local-Hukou migrants in the sectors where unrest increased. The evidence therefore points to a common mechanism operating through the interaction of formalization, limited expansion of employment opportunities, and sector-specific earnings adjustment. Because these channels operate simultaneously, our design identifies their combined effect rather than their separate contributions. Moreover, converter exit from the CMDS sampling frame limits the interpretation of the wage evidence.

In terms of policy, the results challenge the notion that reducing migration barriers necessarily improves labor-market stability. Formal registration changes workers' claims on local institutions, but the consequences depend on how quickly employment opportunities, earnings, and public-service delivery adjust to those new claims. When formal status expands more rapidly than these margins adjust, formalization can generate pressures that manifest in labor conflict rather than immediately improving workers' economic conditions. Identifying the relative contribution of entitlement gaps, employment congestion, and collective-action costs would require

linked worker-firm data that are not available in our city-level design, and so would evaluating more targeted responses. More broadly, the results suggest that the effects of institutional formalization depend not only on who receives formal status, but also on whether the labor market and local institutions can accommodate the claims that status creates.

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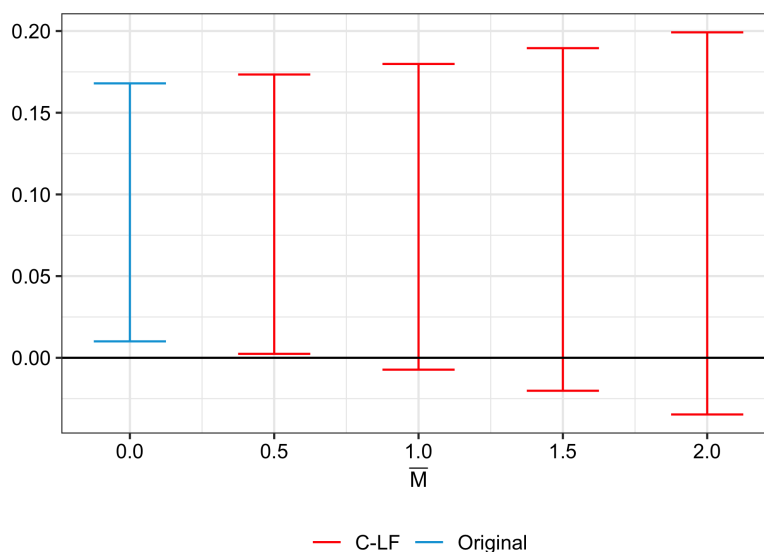
Online Appendix

Does Easing Migration Barriers Increase Labor Unrest? Evidence from the Hukou Reform in China

Unless a note states otherwise, every regression in this appendix compares treatment cities (sizes 4–5, population below 1 million) with control cities (sizes 2–3, population 1–5 million), excludes megacities (population above 5 million), includes city and province-by-year fixed effects, clusters standard errors at the city level, and uses 2013 as the omitted reference year in event studies.

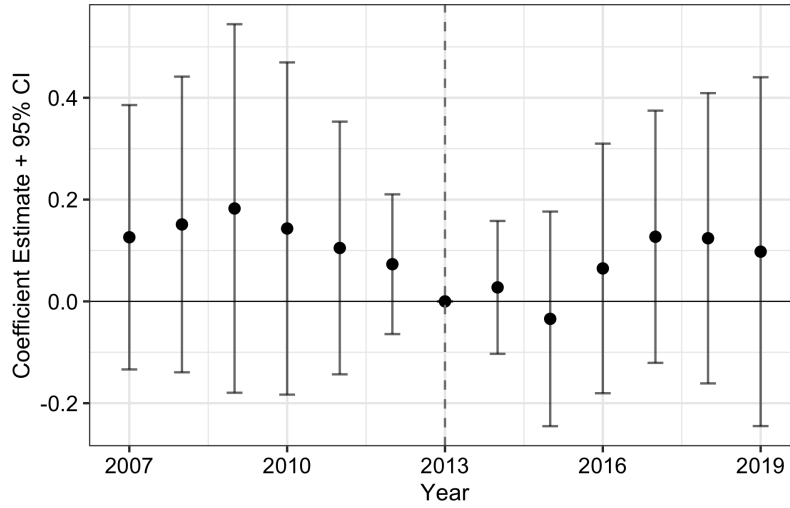
A1 Additional Figures

Figure A1: Sensitivity to Violations of Parallel Trends (2007–2019)



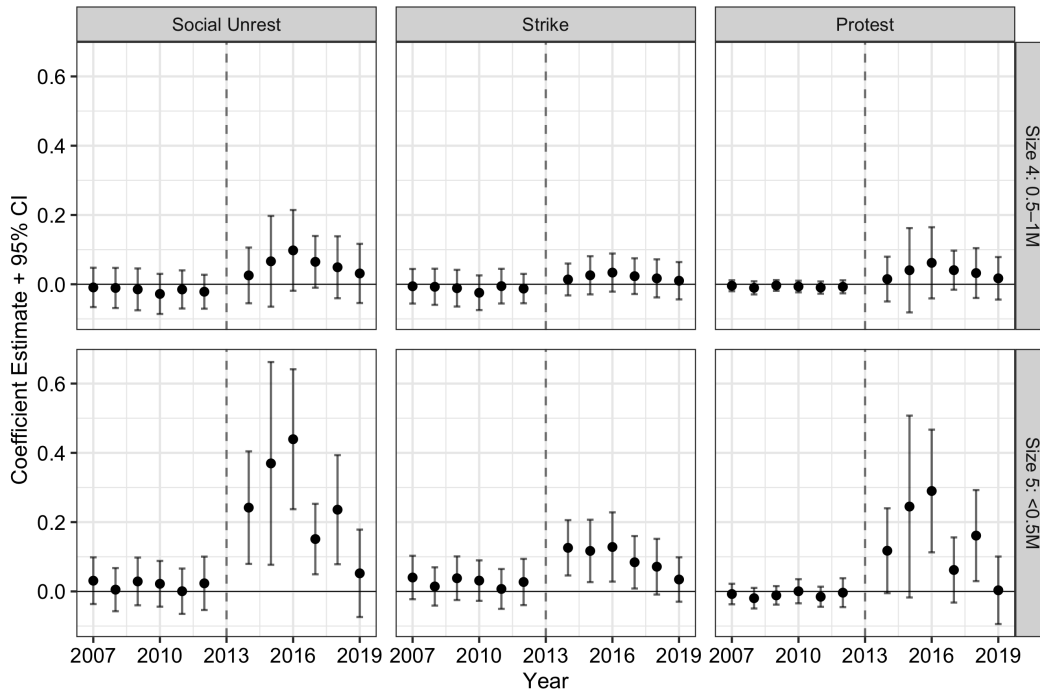
Notes: 95% confidence intervals for the pooled post-treatment effect on per capita overall unrest (events per 100,000 registered population) under the relative magnitudes approach of [Rambachan and Roth \(2023\)](#). Blue interval ($\bar{M} = 0$) assumes exact parallel trends. Red intervals allow post-treatment violations up to $\bar{M}$ times the maximum pre-treatment violation. Six pre-reform coefficients (2007–2012) and six post-reform coefficients (2014–2019).

Figure A2: Event Study, 3 Million Population Cutoff (2007–2019)



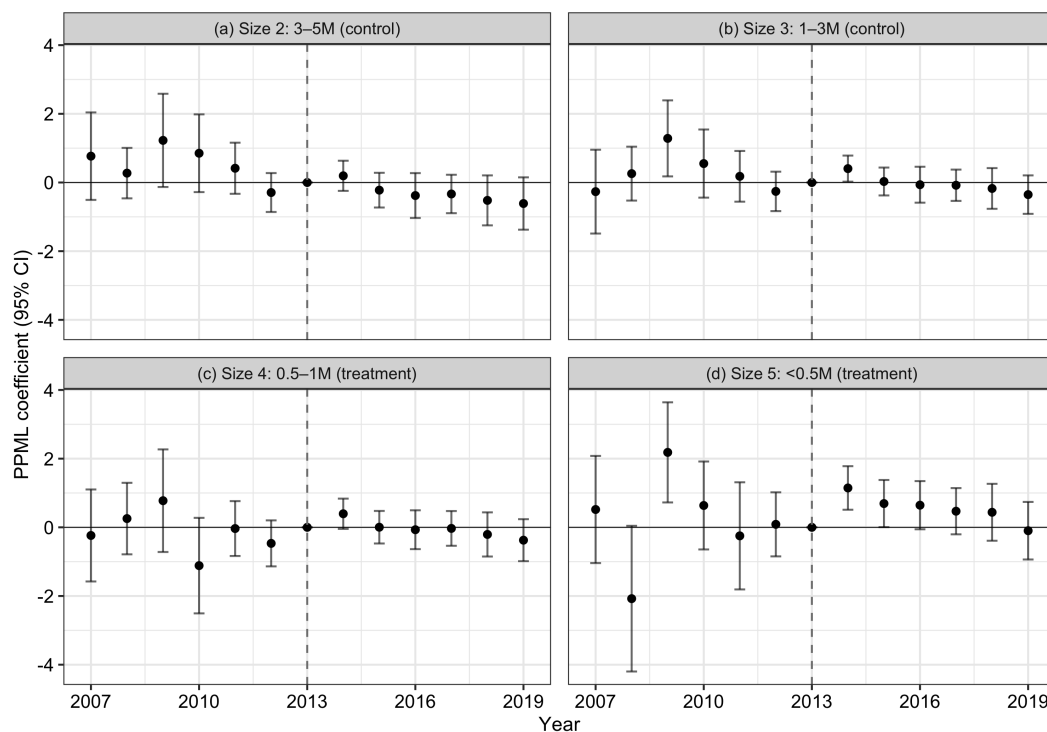
Notes: Event study coefficients and 95% confidence intervals for per capita unrest (events per 100,000 registered population) using a 3 million population cutoff. Treat = 1 for cities with population below 3 million (sizes 3–5, 262 cities). Treat = 0 for cities with population 3–5 million (size 2, 11 cities).

Figure A3: Event Study by City Size, Size 3 as Sole Control (2007–2019)



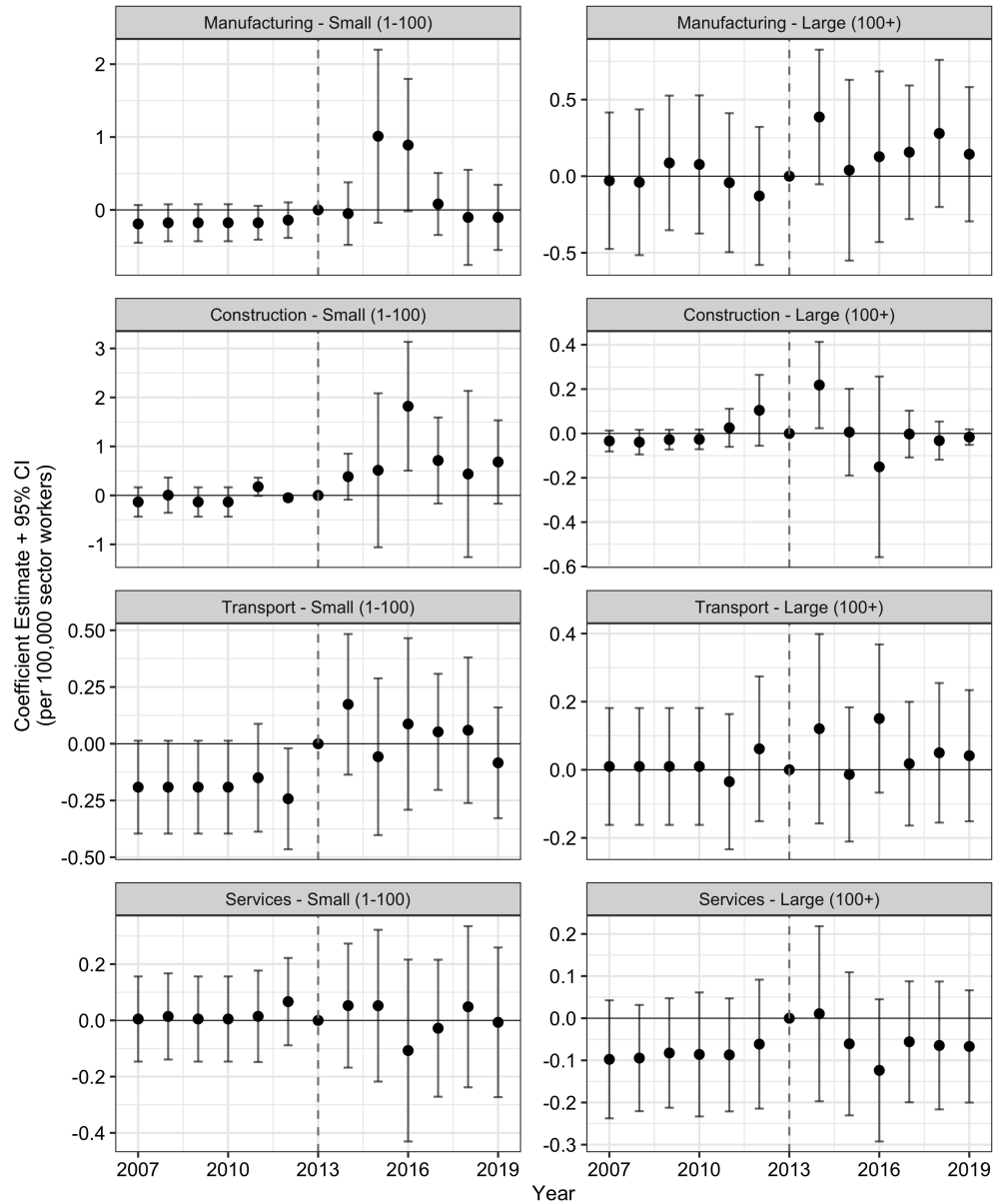
Notes: Event study coefficients and 95% confidence intervals using size 3 cities (population 1–3 million, 108 cities) as the sole control group. Rows show size 4 cities (population 0.5–1 million, 103 cities) and size 5 cities (population below 0.5 million, 51 cities). Columns show overall labor unrest, strikes, and protests, each per 100,000 registered population.

Figure A4: Per Capita Unrest by City Size Category, PPML Estimates (2007–2019)



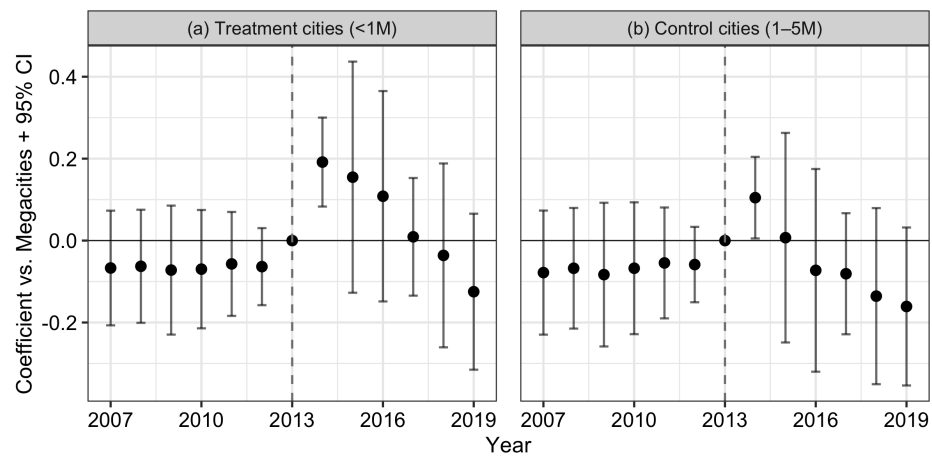
Notes: PPML event-study coefficients and 95% confidence intervals for overall labor unrest per 100,000 registered population, using megacities (size 1, population above 5 million) as the reference group. Panels, sample, fixed effects, clustering, and reference year as in Figure 4. Coefficients are log-multiplicative, so each point measures the proportional differential relative to megacities. The megacity reference group experienced its own protest spike in 2015–2016, which enters every comparison. The wide pre-reform intervals reflect the sparsity of the early records documented in Section 2. Size-5 cities record one event across 51 cities in 2008 and size-4 cities four across 103 in 2010, so a proportional specification is poorly determined in those years.

Figure A5: Unrest by Sector and Scale, Sector-Employment Denominator (2007–2019)



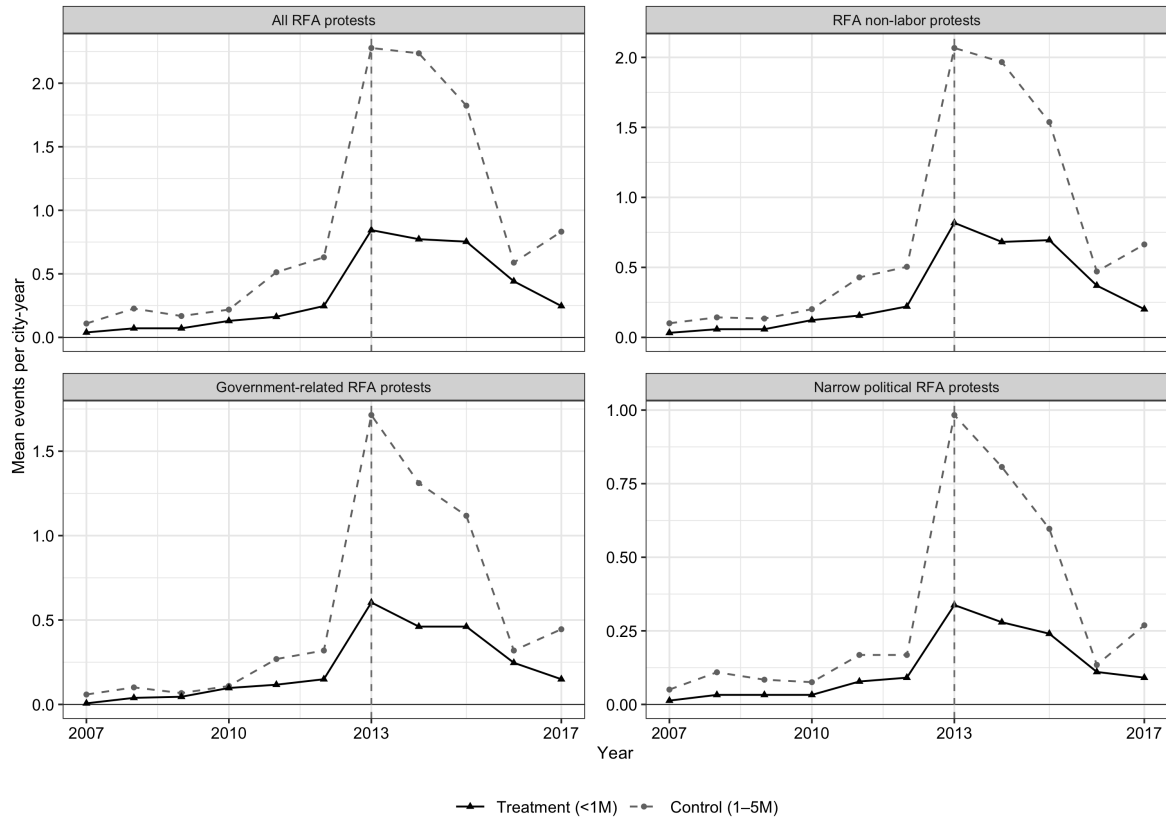
Notes: Event study coefficients and 95% confidence intervals for unrest per 100,000 sector workers. Rows show sectors and columns show incident scale, with sector definitions as in Figure 5. The denominator is secondary-sector employment for manufacturing and construction incidents, and tertiary-sector employment for transport and service incidents. The vertical axis varies across panels.

Figure A6: Per Capita Labor Unrest with Megacities as Control (2007–2019)



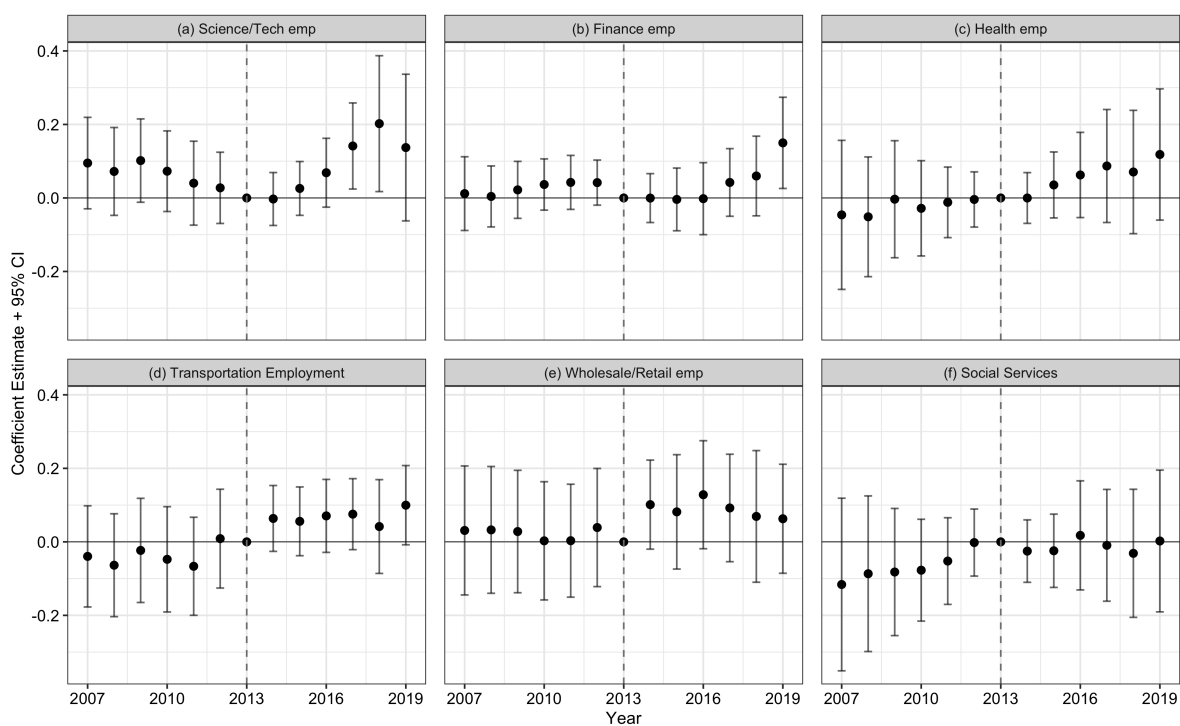
Notes: Event study coefficients and 95% confidence intervals for overall per capita labor unrest (events per 100,000 registered population), using megacities (size 1, population above 5 million) as the reference group. Panel (a) reports treatment cities (sizes 4–5, population below 1 million) and panel (b) reports control cities (sizes 2–3, population 1–5 million). Fixed effects, clustering, and reference year as in Figure 2.

Figure A7: Raw RFA Protest Counts by Treatment Status (2007–2017)



Notes: Raw annual means of separately collected Radio Free Asia (RFA) protest events per city-year. Treatment cities have fewer than 1 million urban residents (sizes 4–5) and control cities 1–5 million (sizes 2–3). Size-1 megacities are excluded. The vertical line marks the 2013 reference year. The raw series is unbalanced in 2008 because all 51 size-5 cities are absent that year. The four outcomes correspond to the panels of Figure 7.

Figure A8: Service Employment by Subsector (2007–2019)



Notes: Event study coefficients and 95% confidence intervals for log employment in six service subsectors. The top row shows (a) scientific and technical services, (b) finance and insurance, and (c) healthcare. The bottom row shows (d) transportation and communication, (e) wholesale and retail trade, and (f) social services.

A2 Additional Tables

Table A1: CLB Incident Counts by Year and City Size (2007–2019)

Year	By City Size					Total	Analysis Sample			
	Size 1	Size 2	Size 3	Size 4	Size 5		Treatment	Control	Treat. % events	Treat. % pop.
2007	14	19	20	12	4	69	16	39	29.1	32.1
2008	17	18	38	13	1	87	14	56	20.0	32.0
2009	4	6	17	5	3	35	8	23	25.8	31.6
2010	27	20	34	4	5	90	9	54	14.3	31.6
2011	31	44	86	29	5	195	34	130	20.7	31.2
2012	82	80	158	45	14	379	59	238	19.9	31.1
2013	117	135	246	105	20	623	125	381	24.7	30.5
2014	201	208	555	265	82	1,311	347	763	31.3	31.4
2015	474	402	1,059	559	157	2,651	716	1,461	32.9	31.5
2016	485	323	1,038	550	184	2,580	734	1,361	35.0	32.2
2017	288	147	449	266	75	1,225	341	596	36.4	32.5
2018	363	191	648	346	105	1,653	451	839	35.0	32.0
2019	317	191	526	251	61	1,346	312	717	30.3	31.8

Notes: Total CLB-recorded labor unrest incidents (strikes and protests combined) by year and city-size category. “Treatment” pools sizes 4–5 (population below 1 million). “Control” pools sizes 2–3 (population 1–5 million). The final two columns report treatment cities’ share of recorded incidents and of registered population within the analysis sample, which excludes megacities.

Table A2: Pre-Reform Sample Characteristics (2007–2013)

	Treatment (<1M)		Control (1–5M)		Diff.
	Mean	SD	Mean	SD	
Registered population (10,000s)	60.30	(21.49)	169.55	(75.77)	-109.25
Per capita GRP (yuan)	41,797.03	(29,175.48)	51,315.42	(39,749.82)	-9,518.39
Secondary employment	53,587.85	(42,668.19)	186,058.87	(230,164.85)	-132,471.01
Tertiary employment	50,820.62	(24,264.15)	159,869.48	(170,202.26)	-109,048.86
Average wage (yuan)	33,274.17	(11,031.75)	35,606.05	(11,311.66)	-2,331.88
Public revenue per capita	3,655.15	(4,002.14)	4,995.76	(5,599.19)	-1,340.62
Public expenditure per capita	6,506.28	(4,696.05)	6,716.60	(5,789.37)	-210.32
Industrial enterprises	189.44	(188.64)	990.43	(1,515.37)	-801.00
Unrest events (count)	0.25	(0.68)	1.11	(4.34)	-0.86
Unrest events per 100k	0.04	(0.11)	0.06	(0.19)	-0.02
Cities	154		119		

Notes: Pre-reform means (2007–2013) for treatment cities (sizes 4–5, population below 1 million) and control cities (sizes 2–3, population 1–5 million). Population is registered (Hukou) population in units of 10,000. Per capita GRP, public revenue, and public expenditure are in yuan per registered resident. “Diff.” reports the difference in means (treatment minus control).

Table A3: Robustness, Classification Using 2014 Population (2007–2019)

	Unrest (1)	Strike (2)	Protest (3)
Treat × Post	0.124*** (0.034)	0.051*** (0.010)	0.071** (0.029)
Observations	3,497	3,497	3,497
R ²	0.64	0.48	0.61

Notes: Treat × Post coefficients for per capita unrest (events per 100,000 registered population) using 2014 population for city size classification instead of the baseline 2013 classification. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A4: Pre-Trends Joint F -Tests (2007–2019)

Specification	F -statistic	p -value	Pre-reform coefficients
<i>Panel A. Per capita rates, OLS</i>			
Unrest/100k	0.74	0.619	6
Strike/100k	0.68	0.665	6
Protest/100k	0.58	0.744	6
<i>Panel B. Event counts, PPML</i>			
Unrest counts	1.03	0.404	6
Strike counts	1.03	0.403	6
Protest counts	0.42	0.866	6
<i>Panel C. Sector and scale subcategories</i>			
Manufacturing, small	0.54	0.747	5
Manufacturing, large	1.48	0.182	6
Construction, small	1.92	0.104	4
Construction, large	0.58	0.748	6
Transportation, small	0.92	0.428	3
Transportation, large	1.03	0.376	3
Services, small	0.55	0.696	4
Services, large	0.59	0.740	6
<i>Panel D. Mechanism outcomes and specification checks</i>			
Registered population	1.53	0.170	6
Secondary employment	1.13	0.348	6
Tertiary employment	0.57	0.754	6
City linear trends	0.67	0.671	6

Notes: Joint F -tests that the estimable pre-reform event-study coefficients (2007–2012) are jointly zero. Panel A covers the three headline per capita outcomes estimated by OLS and Panel B their count PPML counterparts. Panel C covers the eight sector-by-scale categories of Figure 5, with Small = 1–100 participants and Large = 100 or more. Panel D covers the mechanism outcomes and the specification that adds city-specific linear trends. The final column reports how many pre-reform coefficients enter each test. Sparse cells support fewer than six, since the clustered variance matrix is singular in the remaining directions, and the test uses its numerical rank as the degrees of freedom.

Table A5: Robustness, Excluding Transportation Incidents (2007–2019)

	Unrest (1)	Strike (2)	Protest (3)
Treat $\times$ Post	0.076** (0.030)	0.016** (0.006)	0.060** (0.028)
Control post mean	0.339	0.038	0.299
Observations	3,510	3,510	3,510
R ²	0.62	0.49	0.61

Notes: Treat $\times$ Post estimates for per capita unrest rates (events per 100,000 registered residents) after excluding all incidents recorded in transportation, storage, logistics, and postal services, including the taxi and transportation incidents in the 2007–2010 records. Post = 1 for years after 2013. Control post mean is the control-group mean over 2014–2019, the denominator against which the proportional effects in the text are computed. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A6: Robustness, Incidents with 100 or More Participants (2007–2019)

	Unrest (1)	Strike (2)	Protest (3)
Treat $\times$ Post	0.021*** (0.008)	0.015*** (0.006)	0.007 (0.005)
Control post mean	0.069	0.029	0.040
Observations	3,510	3,510	3,510
R ²	0.50	0.45	0.36

Notes: Treat $\times$ Post estimates for per capita unrest rates (events per 100,000 registered residents) counting only incidents with 100 or more recorded participants. Post = 1 for years after 2013. Control post mean is the control-group mean over 2014–2019, the denominator against which the proportional effects in the text are computed. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A7: DiD Estimates for Log Registered Population (2007–2019)

	(Log) Population (1)
Treat $\times$ Post	0.057* (0.033)
Pre-reform mean	4.472
Observations	3,510
R ²	0.95

Notes: Treat $\times$ Post coefficient for log registered population. Post = 1 for years after 2013. Pre-reform mean is the 2007–2013 sample mean of the dependent variable, which is in logs. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A8: Heterogeneity by Industry and Incident Scale, Per Capita Rates (2007–2019)

	Estimate	Control post mean	N
<i>Manufacturing</i>			
Small (1–100 participants)	0.020* (0.012)	0.085	3,510
Large (100+ participants)	0.013*** (0.004)	0.040	3,510
<i>Construction</i>			
Small	0.034** (0.014)	0.132	3,510
Large	0.002 (0.003)	0.007	3,510
<i>Transport</i>			
Small	0.026*** (0.005)	0.031	3,510
Large	0.005* (0.003)	0.013	3,510
<i>Services</i>			
Small	0.001 (0.006)	0.057	3,510
Large	0.002 (0.003)	0.008	3,510

Notes: Each row reports the pooled DiD estimate (Treat $\times$ Post) for per capita unrest (events per 100,000 registered population) in the indicated sector-scale category. Control post mean is the control-group mean over 2014–2019, the denominator against which the proportional effects in the text are computed. Sector and scale definitions as in Figure 5. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A9: Construction Incident Counts by Year, Scale, and Group (2007–2013)

Year	Small (1–100)		Large (100+)	
	Control	Treatment	Control	Treatment
2007	0	0	1	0
2008	0	1	2	1
2009	0	0	1	0
2010	0	0	0	0
2011	2	5	6	2
2012	23	3	9	5
2013	3	1	3	3

Notes: CLB-recorded construction-sector incidents by period, incident scale, and treatment group. Small = 1–100 participants. Large = 100+ participants. Construction events are sparse before electronic CLB coverage begins in 2011. Small-scale construction appears only once in 2007–2010, in treatment cities in 2008.

Table A10: Heterogeneity by Industry and Scale, Per Sector-Worker Rates (2007–2019)

	Estimate	Control post mean	<i>N</i>
<i>Manufacturing</i>			
Small (1–100 participants)	0.436** (0.215)	0.955	3,510
Large (100+ participants)	0.200** (0.094)	0.390	3,510
<i>Construction</i>			
Small	0.795** (0.331)	2.122	3,510
Large	0.003 (0.041)	0.118	3,510
<i>Transport</i>			
Small	0.204*** (0.061)	0.400	3,510
Large	0.052 (0.033)	0.165	3,510
<i>Services</i>			
Small	-0.014 (0.056)	0.612	3,510
Large	0.012 (0.027)	0.098	3,510

Notes: Treat $\times$ Post coefficients for unrest per 100,000 sector workers. Manufacturing and construction use secondary-sector employment as the denominator. Transport and services use tertiary-sector employment. Control post mean is the control-group mean over 2014–2019, the denominator against which the proportional effects in the text are computed. Sector and scale definitions as in Figure 5. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A11: PPML Estimates for Per Capita Unrest Rates (2007–2019)

	Unrest/100k (1)	Strike/100k (2)	Protest/100k (3)
Treat $\times$ Post	0.327** (0.154)	0.532*** (0.175)	0.263 (0.232)
Observations	3,173	3,025	2,495

Notes: Poisson pseudo-maximum likelihood estimates of the primary per capita specification. Coefficients are log-multiplicative, so exact percentage changes are $100[\exp(\hat{\beta}) - 1]$. The OLS counterparts are columns (1)–(3) of Table 1. Sample, fixed effects, and clustering as in Figure 2. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A12: DiD Estimates with Megacities as Control (2007–2019)

	Unrest (1)	Strike (2)	Protest (3)
1–5M $\times$ Post	0.002 (0.090)	0.024* (0.014)	-0.022 (0.089)
<1M $\times$ Post	0.107 (0.092)	0.068*** (0.016)	0.038 (0.088)
Megacity post mean	0.361	0.038	0.321
Observations	3,614	3,614	3,614
R ²	0.64	0.47	0.61

Notes: Treat $\times$ Post coefficients for per capita unrest (events per 100,000 contemporaneous registered residents) using megacities (population above 5 million) as the reference group. “<1M” denotes sizes 4–5. “1–5M” denotes sizes 2–3. Post = 1 for years after 2013. Megacity post mean is the reference group’s mean over 2014–2019, the denominator against which the proportional effects in the text are computed. Because registered population is affected by the reform, these rates describe events per legal resident and are not interpreted as total incident effects. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A13: Alternative Population Cutoffs (2007–2019)

	Unrest (1)	Strike (2)	Protest (3)
<i>Panel A. Treatment = Below 3 Million, Control = 3–5 Million</i>			
<3M $\times$ Post	-0.044 (0.052)	0.045 (0.033)	-0.086 (0.066)
Observations	3,510	3,510	3,510
R ²	0.63	0.47	0.61
<i>Panel B. Treatment = Below 1 Million by Size, Control = 1–3 Million</i>			
0.5–1M $\times$ Post	0.070** (0.033)	0.030*** (0.009)	0.041 (0.028)
<0.5M $\times$ Post	0.233*** (0.060)	0.071*** (0.019)	0.155*** (0.049)
Observations	3,367	3,367	3,367
R ²	0.63	0.44	0.61

Notes: All columns report per capita unrest (events per 100,000 registered population). Each panel re-estimates the pooled DiD with an alternative split of the sample by 2013 urban population. Panel A treats all 262 cities below 3 million as one group and compares them with the 11 cities of 3–5 million. Panel B reports separate effects for cities of 0.5–1 million and cities below 0.5 million, compared with the 108 cities of 1–3 million. The contrast between the panels reflects the design. Panel A pools heavily and barely treated cities and relies on 11 comparison cities, while Panel B aligns the comparison with treatment intensity. Post = 1 for years after 2013. Fixed effects and clustering as in Figure 2. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A14: Migrant Composition Response to Hukou Reform (2011–2017)

	Recent (1)	Low Edu. (2)	Intra-prov. (3)	Any Insur. (4)
Treat $\times$ Post	0.004 (0.008)	-0.0007 (0.005)	0.007 (0.006)	-0.021** (0.008)
Pre-reform mean	0.386	0.700	0.513	0.301
Individual controls	✓	✓	✓	✓
Observations	485,496	485,496	485,496	196,046
R ²	0.06	0.21	0.26	0.29

Notes: Each column reports the pooled DiD estimate (Treat $\times$ Post) for the indicated migrant characteristic. Recent = 1 if arrived within 3 years. Low education = 1 if 9 or fewer years of schooling. Any insurance = 1 if holding any social insurance. Individual controls include gender, marital status, and Hukou type. City, province-by-year, and occupation-by-year fixed effects. Standard errors clustered at the city level. Data from the CMDS. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A15: DiD Estimates for Sectoral Employment (2007–2019)

Panel A. Broad Sectors

	Secondary (1)	Tertiary (2)
Treat × Post	-0.041 (0.046)	0.041* (0.022)
Observations	3,510	3,510
R ²	0.95	0.97

Panel B. Service Subsectors

	SciTech (1)	Finance (2)	Health (3)	Transport (4)	Retail (5)	SocialServ (6)
Treat × Post	0.037 (0.064)	0.018 (0.036)	0.083 (0.081)	0.101* (0.053)	0.070 (0.056)	0.048 (0.078)
Observations	3,510	3,510	3,216	3,510	3,510	3,212
R ²	0.91	0.92	0.88	0.92	0.90	0.92

Notes: Treat $\times$ Post coefficients for log employment. Panel A reports secondary and tertiary sectors. Panel B decomposes tertiary employment into six subsectors. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A16: CMDS Migrant Sample Summary Statistics (2011–2017)

	Treatment (<1M)		Control (1–5M)	
	Pre	Post	Pre	Post
Log monthly wage	7.854 (0.553)	8.056 (0.575)	7.869 (0.507)	8.150 (0.525)
Age	35.687 (8.822)	36.763 (9.662)	34.836 (8.589)	36.084 (9.432)
Male	0.549 (0.498)	0.551 (0.497)	0.522 (0.500)	0.538 (0.499)
Low education (≤ 9 yrs)	0.753 (0.431)	0.695 (0.461)	0.704 (0.456)	0.650 (0.477)
Married	0.839 (0.368)	0.848 (0.359)	0.831 (0.375)	0.839 (0.368)
Any social insurance	0.173 (0.378)	0.147 (0.354)	0.273 (0.446)	0.259 (0.438)
Recent migrant (≤ 3 yrs)	0.373 (0.484)	0.299 (0.458)	0.386 (0.487)	0.334 (0.472)
Manufacturing	0.137 (0.344)	0.143 (0.350)	0.246 (0.431)	0.234 (0.423)
Construction	0.098 (0.297)	0.078 (0.268)	0.082 (0.275)	0.074 (0.261)
Transportation	0.126 (0.332)	0.139 (0.346)	0.121 (0.326)	0.131 (0.338)
Observations	51,546	97,822	108,148	206,798

Notes: Summary statistics from the China Migrants Dynamic Survey (CMDS), 2011–2017. Treatment cities have population below 1 million (sizes 4–5). Control cities have population 1–5 million (sizes 2–3). “Pre” covers 2011–2013 and “Post” covers 2014–2017. Low education is defined as 9 or fewer years of schooling. Recent migrant is defined as having arrived in the current city within the past 3 years. Standard deviations in parentheses.

Table A17: Robustness, CMDS Migrant Results (2011–2017)

	All (1)	Manufacturing (2)	Construction (3)	Transportation (4)	Other services (5)
<i>Panel A. Migrant Wage Robustness</i>					
(a) Baseline	-0.037*** (0.009)	-0.017 (0.012)	-0.041*** (0.015)	-0.038** (0.015)	-0.041*** (0.012)
(b) No Occ×Year FE	-0.045*** (0.010)	-0.021 (0.013)	-0.037** (0.016)	-0.039** (0.015)	-0.051*** (0.012)
(c) No Hukou control	-0.037*** (0.009)	-0.017 (0.012)	-0.040*** (0.015)	-0.038** (0.015)	-0.041*** (0.012)
Pre-reform mean	7.885	7.876	8.028	7.777	7.892
Observations	424,723	99,093	36,883	59,106	229,622
<i>Panel B. Sector Choice Robustness</i>					
(a) Baseline		0.003 (0.004)	-0.004 (0.002)	-0.001 (0.002)	0.017*** (0.005)
(b) No Occ×Year FE		0.006 (0.006)	-0.007* (0.004)	0.004 (0.004)	0.015** (0.008)
(c) No Hukou control		0.003 (0.004)	-0.004 (0.002)	-0.001 (0.002)	0.017*** (0.005)
Pre-reform mean		0.246	0.082	0.121	0.441
Observations		485,496	485,496	485,496	485,496

Notes: Panel A reports the Treat × Post coefficient with log monthly wage as the dependent variable. Panel B uses a sector indicator as the dependent variable. Row (a) is the baseline from Table 2. Row (b) drops occupation-by-year FE. Row (c) drops the Hukou type control. All specifications include individual controls, city FE, and province-by-year FE. Standard errors clustered at the city level. Data from the CMDS (2011–2017). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A18: Migrant Wage Effects by Tenure (2011–2017)

	Manufacturing (1)	Construction (2)	Transportation (3)	Other services (4)
Treat × Post	-0.021 (0.014)	-0.037** (0.017)	-0.032* (0.018)	-0.034** (0.013)
Treat × Post × Recent	0.017 (0.019)	-0.006 (0.022)	-0.014 (0.021)	-0.020 (0.013)
Pre-reform mean	7.876	8.028	7.777	7.892
Observations	99,093	36,883	59,106	229,622
R ²	0.36	0.27	0.26	0.24

Notes: Each column reports the Treat × Post coefficient and its interaction with a Recent indicator (arrived within 3 years). Columns restrict to migrants in the indicated sector. Individual controls, city, province-by-year, and occupation-by-year fixed effects. Standard errors clustered at the city level. Data from the CMDS (2011–2017). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A19: Economic and Fiscal Responses (2007–2019)

	Average Wage (1)	Revenue (2)	Expenditure (3)
Treat $\times$ Post	-0.006 (0.014)	-0.020 (0.039)	0.006 (0.037)
Observations	3,510	3,510	3,510
R ²	0.95	0.96	0.95

Notes: Treat $\times$ Post coefficients for log outcomes. Columns report (1) log average wages of employed persons, (2) log total local public revenue, and (3) log total local public expenditure, the latter two before division by registered population. Post = 1 for years after 2013. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.